ENTERPRISE CREDIT DECISIONING STRATEGY SIMULATOR

Module 1: Synthetic Application & Risk Modeling Engine v2.0

Validation Summary (Board-Ready)

1. EXECUTIVE SUMMARY

The Module 1 Synthetic Application & Risk Modeling Engine v2.0 has been successfully validated as a **portfolio-grade, deterministic, and scenario-capable simulation framework** designed to emulate real-world credit decisioning environments without exposing sensitive data.

Building on the validated foundation established in v1.0, this release introduces **targeted architectural and behavioral enhancements across four critical workstreams**—mortgage realism, estimated-PD proxy dispersion, scenario framework design, and revolving credit sensitivity—resulting in a more **robust, comparable, and decision-ready simulation environment**.

This validation confirms that the engine:

- Produces **stable, fully reproducible populations** using deterministic seeding tied to population_id, enabling consistent cross-scenario comparison
- Accurately reflects configured portfolio composition across product types and credit score segments, with **no degradation from prior validated baselines**
- Generates **behaviorally realistic borrower profiles**, preserving meaningful overlap across risk dimensions while avoiding artificial segmentation
- Maintains **coherent and economically intuitive financial relationships** between income, exposure, payment burden, affordability (PTI), and downstream estimated risk
- Demonstrates a **directionally correct, explainable, and now better-dispersed estimated-PD proxy framework**, where estimated risk and Expected Loss (EL) respond appropriately to borrower characteristics and structural loan features

- Supports **matched, row-level scenario comparison**, enabling controlled evaluation of strategy changes without population noise

Key V2.0 validation outcomes include:

- **Mortgage exposure behavior corrected and stabilized.** Across the Workstream A sequence, maximum requested amount declined from approximately **$1.15M in V1.0** to approximately **$1.05M in final A3**, while mortgage median requested amount remained stable at approximately **$293K**. This confirms that tail behavior was controlled without collapsing the core mortgage distribution.
- **Estimated-PD proxy distribution refined.** Workstream B reduced capped estimated-PD proxy observations from **8,861 rows / 17.72% in B1** to **6,428 rows / 12.86% in final B3**, reducing mechanical upper-bound compression while preserving directional score-band ordering.
- **Scenario framework successfully implemented and validated.** Scenario archive outputs retained **50,000 rows per scenario**, used the same `population_id`, and supported matched row-level comparison across baseline and rate-stress configurations.
- **Revolving payment dynamics enhanced.** In the final rate-up scenario, revolving payment deltas became positive across score bands, confirming that APR movement now flows into payment burden, PTI, estimated-PD proxy, and Expected Loss.

Residual edge behaviors were reviewed and determined to be **intentional, bounded, and representative of real-world portfolio variability**, with no evidence of instability or structural breakdown.

Overall, Module 1 v2.0 has achieved its objective:

To function as a **governed, explainable, scenario-capable, and portfolio-realistic simulation engine** suitable for pre-production credit strategy testing, sensitivity analysis, and decision framework design.

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2. VALIDATION APPROACH OVERVIEW

2.1 Validation Framework Overview

- Validation was conducted using a **structured, multi-layer QA framework embedded directly within the SQL implementation (Section 9)**. This framework evaluates the simulation across five core dimensions:
- **Structural Integrity** — correctness of portfolio construction and parameter adherence
- **Distribution Alignment** — consistency between configured assumptions and observed outputs
- **Behavioral Realism** — plausibility of borrower profiles and cross-variable relationships
- **Risk Model Coherence** — directional correctness and logical consistency of the estimated-PD proxy, LGD assumptions, and Expected Loss
- **Edge Case & Overlap Validation —** presence and control of realistic outliers and mixed-signal scenarios
- Each dimension includes both:
- **Aggregate statistical validation** to confirm portfolio-level alignment
- **Targeted diagnostic queries** to support interpretability, anomaly detection, and root-cause analysis

Because Module 1 uses a synthetic estimated-PD proxy rather than an empirically calibrated production model, validation emphasizes directional coherence, bounded behavior, explainability, and scenario sensitivity rather than calibration to observed default rates.

2.2 V2.0 Validation Enhancements

In addition to the foundational framework established in V1.0, validation in V2.0 introduces an expanded validation approach focused on **workstream-based evidence, scenario consistency, tail-frequency review, and matched comparison**.

Workstream-Based Validation Tracks

Each major system enhancement—mortgage realism, estimated-PD proxy dispersion, scenario framework design, and revolving sensitivity—is validated independently using targeted QA outputs, diagnostic evidence, and outcome analysis.

This ensures that each V2.0 enhancement is evaluated on its own merits while also confirming that no workstream destabilizes the broader portfolio structure.

Scenario Consistency Validation

The newly introduced scenario framework is validated to ensure:

- Identical borrower populations across scenario runs
- Stable outputs under identical inputs
- Controlled variation driven by parameter changes
- Matched row-level comparison across baseline and rate-stress configurations

This confirms that observed scenario differences are attributable to controlled parameter changes rather than stochastic population variation.

Tail-Frequency Validation

V2.0 expands tail validation beyond percentile and maximum-value review by explicitly measuring the frequency of high-affordability-stress observations, including:

- PTI > 40%
- PTI > 50%
- PTI > 75%

This confirms that high-PTI observations remain **present but non-dominant**, supporting realistic portfolio variation without allowing extreme affordability stress to distort aggregate behavior.

Diagnostic Transparency

V2.0 validation also incorporates diagnostic outputs for mortgage affordability refinement and estimated-PD proxy saturation behavior.

These diagnostics support:

- Review of intermediate mortgage transformation logic
- Evaluation of PTI-aware dampening behavior
- Measurement of estimated-PD proxy cap crowding
- Confirmation that high-risk observations remain differentiated below the governance cap

Together, these enhancements extend validation beyond static correctness to include **comparability, stability, sensitivity, tail governance, and diagnostic explainability**.

2.3 Validation Methodology

Validation is performed using a combination of:

- **Portfolio-level statistical analysis**
  Confirms row counts, product mix, score-band mix, and key distributional statistics across baseline and workstream iterations.
- **Regression-control validation**
  Confirms that workstream changes did not unintentionally alter population size, product mix, score-band composition, or borrower attribute generation.
- **Segment-level diagnostics**
  Reviews product, score-band, PTI-bucket, and product × score-band summaries to confirm directional realism and identify compression or tail issues.
- **Workstream iteration evidence**
  Documents the progression from initial issue to final accepted design across Workstreams A and B, including A1 → A2 → A3 and B1 → B2 → B3.
- **Matched scenario comparison**
  Compares identical borrower populations across archived scenarios to isolate parameter-driven effects from population noise.
- **Edge-case review**
  Confirms that mixed-signal cases remain present, including strong-credit/high-PTI, weaker-credit/manageable-PTI, and large-exposure/manageable-affordability examples.
- **Tail-frequency validation**
  Quantifies the prevalence of high-affordability-stress observations, including PTI >40%, PTI >50%, and PTI >75%, to confirm that tail behavior remains present but non-dominant.

This layered approach ensures that validation captures not only **correctness**, but also **economic realism, internal consistency, and decisioning relevance**.

2.4 Validation Philosophy

The objective of validation is not to confirm that the system produces idealized outputs, but to ensure that it produces **controlled, explainable, and economically coherent behavior** under a wide range of conditions.

Emphasis is placed on:

- **Transparency over opacity**
- **Deterministic stability over stochastic variability**
- **Directional correctness over overfitting or artificial precision**

This approach ensures the engine functions as a **governed simulation environment**, suitable for strategy evaluation rather than predictive deployment.

3. PORTFOLIO STRUCTURE VALIDATION

3.1 Product Mix Alignment

Objective: Confirm that generated product distribution matches configured portfolio assumptions and remains stable under the V2.0 architecture.

Observed Output

Product Type	Applications	% of Total
Mortgage	15,065	30.13%
Revolving Line	12,455	24.91%
Auto Loan	9,946	19.89%
Unsecured Personal Loan	7,572	15.14%
HELOC	4,962	9.92%

Assessment

- Target mix: **30 / 25 / 20 / 15 / 10**
- Observed deviation: **< 0.2% across all categories**
- No evidence of drift or instability introduced by V2.0 enhancements

Conclusion

Product assignment logic remains **highly precise, stable, and fully controlled**. The final V2.0 population preserved the configured product mix across **50,000 applications**, with observed shares matching the intended 30 / 25 / 20 / 15 / 10 design within approximately **0.2 percentage points across all product categories**. This confirms that downstream workstream enhancements did not distort foundational portfolio construction.

3.2 Score Band Distribution

Objective: Validate that score band distribution remains balanced, well-shaped, and suitable for downstream segmentation and risk analysis.

Observed Output

Score Band	Applications	% of Total
Super Prime	8,984	17.97%
Prime	13,987	27.97%
Near Prime	12,073	24.15%
Subprime	8,902	17.80%
Deep Subprime	6,054	12.11%

Assessment

- Balanced population across all score bands
- No collapse into extreme segments (e.g., over-concentration in subprime)
- Adequate density within each band to support statistical analysis and scenario testing
- Distribution remains consistent with configured assumptions
- No distortion was introduced by V2.0 enhancements. The same portfolio structure was preserved across Workstream A, Workstream B, and scenario-framework validation, confirming that observed changes in mortgage affordability, estimated-PD proxy dispersion, and scenario outcomes were attributable to intended model changes rather than population drift.

Conclusion

The engine continues to produce a **well-formed, analytically usable portfolio**, supporting meaningful segmentation and risk differentiation.

3.3 V2.0 Stability Confirmation

Objective: Confirm that structural enhancements introduced in V2.0 do not degrade foundational portfolio construction behavior.

Validation Focus

- Product assignment stability
- Score band integrity
- Population balance under deterministic generation

Findings

- No measurable drift in product or score distributions
- Deterministic population generation ensures identical structure across runs
- Scenario framework preserves population consistency across scenario variations

Conclusion

V2.0 enhancements are **non-destructive to core portfolio structure**, preserving all foundational behaviors validated in V1.0.

3.4 Interpretation

Portfolio structure validation confirms that the system maintains:

- **Strict adherence to configured assumptions**
- **Stable and reproducible population composition**
- **A reliable foundation for downstream behavioral and risk validation**

This establishes confidence that subsequent validation results are driven by **intended system behavior**, not structural inconsistencies.

4. DETERMINISTIC GENERATION VALIDATION

4.1 Deterministic Design Validation

Objective

- Validate that the system produces **fully deterministic outputs**, ensuring identical results for identical inputs.

Implementation Overview

- The engine replaces session-based randomness (e.g., random()) with deterministic hashing:
- MD5(application_id + population_id + deterministic_key_suffix)
- Outputs are transformed into uniform(0,1) values and used to drive all stochastic components.

Assessment

- No reliance on session-level randomness
- All variable generation anchored to deterministic inputs
- Reproducibility guaranteed at the row level

Conclusion

- The system satisfies deterministic design requirements, functioning as a **controlled, reproducible simulation environment**.

4.2 Reproducibility Validation Results

Test Design

Multiple runs executed with identical:

- population_id
- parameter configurations
- deterministic seed structure

Observed Outcome

Reproducibility validation confirmed **100% identical outputs across repeated runs using identical population and configuration controls**. No variation was observed in:

- Product assignment
- Borrower attributes
- Pricing variables
- Payment variables
- PTI
- Estimated-PD proxy
- Expected Loss

Conclusion

The engine is **fully reproducible under identical inputs**, supporting stable QA, audit traceability, debugging, and matched scenario comparison.

4.3 Scenario Comparability Validation

Objective

Validate that the scenario framework produces **directly comparable outputs** by preserving identical borrower populations across scenarios while allowing controlled variation driven by parameter changes.

Validation Design

Multiple scenarios were executed within the same `scenario_set_name` using:

- A fixed `population_id`
- Identical borrower generation inputs
- Controlled variation in selected parameters (e.g., `rate_shift_bps`)

Outputs were compared at the **row level** across scenarios.

Observed Results

Population Consistency

- **50,000 baseline rows and 50,000 rate-stress rows** were produced
- All rows shared the same `population_id`
- **100% of borrower-level attributes** (credit score, income, DTI, utilization) remained identical across scenarios

This confirms:

Population generation is fully deterministic and preserved across scenario runs

Controlled Outcome Variation

In the validated 200BP rate-up scenario:

Metric	Baseline	200BP Rate-Up Scenario	Change
Avg APR	~14.56%	~16.14%	+1.58 pts
Avg Monthly Payment	~$1,511	~$1,612	+$101
Avg PTI	~16.50%	~17.41%	+0.91 pts

This confirms:

Parameter changes propagate through pricing → payment → affordability as designed

Row-Level Sensitivity

- **~6.7% of observations (3,367 / 50,000)** exhibited an increase in estimated-PD proxy
- Changes were **not uniform** across the population
- Increases were concentrated in:
    - Higher-risk segments
    - Higher-PTI observations

This confirms:

Scenario effects are **targeted and economically coherent**, not mechanically applied

Interpretation

Scenario comparability validation confirms that:

- The framework preserves **identical borrower populations across runs**
- Output differences are **fully attributable to parameter changes**
- The modeled transmission path (APR → Payment → PTI → estimated-PD proxy) is preserved under scenario variation.
- Scenario impacts are **directionally correct and segment-sensitive**, not uniformly distributed

Conclusion

The scenario framework enables **true matched-sample comparison**, where:

- Population consistency is guaranteed
- Parameter-driven effects are isolated
- Outcome differences are **intentional, explainable, and analytically meaningful**

This establishes the system as a **controlled experimentation environment**, suitable for strategy testing, sensitivity analysis, and decision framework design.

4.4 Stability & Debug Traceability

Findings

Deterministic generation enables:

- **Stable QA outputs**, eliminating run-to-run noise
- **Repeatable validation results**, supporting regression testing
- **Precise debugging**, allowing variable-level traceability back to deterministic inputs

Conclusion

The system supports **robust validation workflows and audit requirements**, significantly reducing complexity in model review and issue resolution.

4.5 Interpretation

Deterministic generation is not a technical convenience—it is a **foundational requirement** for:

- Scenario-based analysis
- Controlled experimentation
- Reliable validation and governance

By eliminating stochastic variation, the system ensures that:

All observed differences in outputs are intentional, explainable, and attributable to defined inputs.

5. VARIABLE DISTRIBUTION VALIDATION

5.1 Key Portfolio Statistics (Selected)

Objective: Validate that core variables exhibit **realistic distributions, controlled spread, and stable behavior** following V2.0 enhancements.

Variable	Median	Mean	P75	Max
Annual Income	~$118K	~$122K	~$163K	~$240K
Requested Amount	~$47K	~$133K	~$184K	~$1.05M
PTI	~11.3%	~16.5%	~26.3%	~133% (controlled tail)
Estimated-PD proxy	~8.8%	~16.3%	~31.8%	~45% (bounded)

Metric	V1.0 Baseline	Final V2.0	Validation Interpretation
Max requested amount	~$1.15M	~$1.05M	Upper exposure tail reduced
Max monthly payment	~$10,276	~$9,238	Payment burden tail reduced
Median requested amount	~$47K	~$47K	Core distribution preserved
Median PTI	~11.3%	~11.3%	Central affordability stable
Mean estimated-PD proxy	~16.8%	~16.3%	Estimated risk proxy modestly reduced
Estimated-PD proxy governance cap	45%	45%	Simulation guardrail retained

All statistics are based on a deterministic population of 50,000 synthetic applications. The estimated-PD proxy values are synthetic risk outputs from the Module 1 multiplier framework and should be interpreted as governed simulation measures rather than calibrated production default forecasts.

Tail Behavior Improvement Summary (Baseline / Prior Evidence → V2.0)

To complement distributional statistics, tail-frequency validation quantifies the share of the portfolio in high-affordability-stress conditions. This confirms whether extreme PTI observations remain present but non-dominant, and whether V2.0 improved tail behavior without artificially eliminating realistic edge cases.

Metric	Prior / Baseline Evidence	Final V2.0 Evidence	Change	Validation Interpretation
PTI > 40%	4,081 rows / **8.16%**	3,877 rows / **7.75%**	**-204 rows / -0.41 pts**	High affordability-stress frequency modestly reduced
PTI > 50%	883 rows / **1.77%**	600 rows / **1.20%**	**-283 rows / -0.57 pts**	Severe affordability-stress frequency materially reduced
PTI > 75%	119 rows / **0.24%**	119 rows / **0.24%**	**No change**	Extreme tail remains rare and preserved
Estimated-PD capped at governance maximum	8,861 rows / **17.72%**	6,428 rows / **12.86%**	**-2,433 rows / -4.86 pts**	Upper-bound compression materially reduced
Max monthly payment	**~$10.3K**	**~$9.2K**	**~-$1.0K**	Upper-tail payment burden reduced
Max requested amount	**~$1.15M**	**~$1.05M**	**~-$96K**	Upper-tail exposure controlled

This evidence shows that V2.0 improved tail behavior across both affordability and risk dimensions. High-PTI observations remain present, which is appropriate for a realistic synthetic portfolio, but the severe affordability-stress segment declined from **1.77% to 1.20%** of the population. At the same time, the extreme PTI tail remained rare and unchanged at **0.24%**, indicating that V2.0 did not rely on arbitrary clipping or unrealistic removal of edge cases.

In parallel, Workstream B reduced **estimated-PD proxy cap crowding** from **17.72% to 12.86%**, while Workstream A reduced upper-tail payment and exposure levels. Together, these results confirm that V2.0 strengthened tail governance across both affordability and risk dimensions without collapsing realistic portfolio variation.

5.2 Distribution Shape & Behavior

Findings

- **Income**
  - Right-skewed distribution
  - Strong overlap across score segments
  - No artificial clustering
- **Requested Amount**
  - Product-conditioned distribution
  - Mortgage tail remains present but **controlled (Workstream A)**
  - No runaway exposure scaling
- **PTI (Affordability)**
  - Central tendency within realistic ranges
  - High-PTI tail exists but is **bounded and non-dominant**
  - Improved stability vs V1.0 (reduced extreme inflation)
- **Estimated-PD Proxy**
  - Directionally coherent distribution across the population
  - Reduced clustering near the upper governance cap
  - Improved differentiation in weaker segments **(Workstream B)**
  - The estimated-PD proxy distribution improved in the high-risk tail, as evidenced by the reduction in capped observations from **17.72% to 12.86%.** This reduced mechanical compression near the upper bound and improved differentiation among weaker-credit observations while preserving score-band risk ordering.

5.3 Tail Behavior Validation

Objective

Validate that extreme values exist where appropriate but remain **controlled, bounded, and non-dominant** within the portfolio.

Findings

High PTI Observations (>40%)

- High-PTI observations (>40%) represent a **small minority of the portfolio (single-digit percentage range)** and are concentrated in structurally high-burden scenarios rather than broadly distributed.
    - V2.0 reduced the structural drivers of excessive tail inflation.
- **Workstream A (Final A3) evidence:**
    - Maximum requested amount reduced from approximately **$1.15M → $1.05M**
    - Maximum monthly payment reduced from approximately **$10.3K → $9.2K**
    - Median mortgage requested amount remained stable at approximately **$293K**, confirming preservation of the core distribution
- High-PTI cases are:
    - **Present but non-dominant**
    - **No longer structurally inflated by exposure scaling**
    - Still available for **validation, stress testing, and manual-review realism**

Large Exposure (Mortgage Tail)

- Large exposure observations continue to exist
- However:
    - Exposure is now **tied to income capacity and affordability constraints**
    - Upper-tail behavior is **dampened rather than clipped**

This confirms:

Tail behavior is governed through **affordability-aware logic**, not arbitrary limits

High Estimated-PD Proxy Values

These values should be interpreted as synthetic risk proxy outputs, not calibrated default forecasts.

- Estimated-PD proxy tail behavior improved through Workstream B:
    - Capped observations reduced from:
        - **8,861 rows (17.72%) in B1**
        - to **6,428 rows (12.86%) in final B3**
- Additional observations:
    - Reduced clustering near the 45% cap
    - Gradual tapering observed in high-risk segments
    - Score-band monotonicity preserved

This confirms:

High-risk observations remain differentiated rather than compressed into a single capped value

Conclusion

Tail behavior in V2.0 is:

- **Realistic** — extreme cases exist where expected
- **Governed** — upper-tail exposure, payment, and estimated-PD proxy are controlled through model design
- **Non-dominant** — tail observations do not distort portfolio-level behavior
- **Non-destabilizing** — improvements reduced artificial inflation without collapsing valid edge cases

As a result, the system reflects **real-world portfolio variability** while avoiding structural artifacts introduced by uncontrolled tail behavior.

5.4 V2.0 Distribution Improvements

Compared to V1.0, the following improvements are observed:

- **Mortgage exposure behavior stabilized**
    - Reduced unrealistic PTI inflation
    - Improved affordability alignment
- **The estimated-PD proxy distribution was materially improved and measurably de-compressed near the upper governance bound.**
    - Capped observations declined from **8,861 rows / 17.72% in B1** to **6,428 rows / 12.86% in B3**, reducing mechanical cap compression while preserving the intended monotonic increase in risk from stronger to weaker score bands.
- **Cross-variable coherence strengthened**
    - Stronger alignment between:
        - exposure
        - payment
        - affordability
        - estimated risk

These improvements are complementary: Workstream A stabilizes the affordability inputs driving risk, while Workstream B improves how that risk is distributed across the population.

5.5 Interpretation

Variable distribution validation confirms that:

- The system produces **economically plausible, well-shaped distributions**
- Variables behave as **interconnected components**, not independent random draws
- Structural improvements in V2.0 enhance realism **without introducing instability**

This establishes confidence that downstream analyses reflect **true modeled behavior**, not distributional artifacts.

6. PRODUCT-LEVEL BEHAVIOR VALIDATION

6.1 Objective

Product-level validation confirms that structurally distinct product designs produce **quantitatively and directionally consistent differences in exposure, pricing, payment, affordability, and risk**, as shown below:

6.2 Mortgage Behavior Validation (Workstream A)

Observed Behavior

- Large exposure levels
- PTI concentrated in moderate range (~25–35%)
- Lower APR relative to unsecured products

V2.0 Validation Enhancements

- High-PTI inflation corrected
- Exposure now constrained by income and affordability

Metric	V1.0 Mortgage	Final V2.0 A3 Mortgage	Interpretation
Avg requested amount	~$344K	~$339K	Tail reduced without collapsing product
Median requested amount	~$293K	~$293K	Core distribution preserved
Avg monthly payment	~$3,017	~$2,973	Payment burden improved
Median monthly payment	~$2,609	~$2,604	Central payment behavior stable
Avg PTI	~29.82%	~29.67%	Slight affordability improvement
Max portfolio requested amount	~$1.15M	~$1.05M	Upper-tail exposure controlled
Max portfolio payment	~$10.3K	~$9.2K	Upper-tail payment burden controlled

Conclusion

Workstream A did not eliminate mortgage tail behavior; it governed it. The final A3 iteration reduced upper-tail exposure and payment burden while preserving the core mortgage distribution, confirming that the fix **improved realism without overcorrecting the product.**

6.3 Revolving Credit Behavior Validation (Workstream D)

Observed Behavior

- Smaller balances / limit proxies
- Higher APR ranges
- Payment levels respond to APR changes

Key Validation Check

- Positive relationship observed:

APR ↑ → Payment ↑ → PTI ↑ → estimated-PD proxy ↑

In the prior design, revolving payment did not respond to APR movement. Under Workstream D, this relationship is restored, as shown below:

Revolving Score Band	Avg APR Delta	Avg Payment Delta	Avg PTI Delta
Super Prime	+2.00%	+$15.84	+0.12 pts
Prime	+2.00%	+$15.00	+0.14 pts
Near Prime	+2.00%	+$14.25	+0.16 pts
Subprime	+2.00%	+$13.66	+0.21 pts
Deep Subprime	+1.58%	+$9.12	+0.20 pts

Conclusion

Workstream D corrected the prior limitation in which revolving payment did not respond to APR movement. In the final D1 validation, revolving payment deltas became positive across all score bands, confirming that rate movement now flows through payment burden into PTI and downstream estimated risk. This demonstrates directional APR-sensitive payment behavior while preserving the boundary that the revolving payment proxy is not a full billing-cycle model.

6.4 Auto Loan Behavior Validation

Observed Behavior

- Mid-range exposure
- Structured amortization
- Moderate PTI

Conclusion

Auto loans exhibit **stable, structure-driven affordability behavior**, consistent with expectations.

6.5 Personal Loan Behavior Validation

Observed Behavior

- Smaller balances
- Higher APR relative to secured products
- Higher PTI sensitivity

Conclusion

Personal loans behave as **higher-cost, affordability-sensitive products**, reinforcing structural differentiation.

6.6 HELOC Behavior Validation

Observed Behavior

- Larger exposure with interest-only structure
- Lower immediate PTI vs mortgage
- Distinct from both installment and revolving unsecured

Conclusion

HELOC behavior provides **meaningful contrast**, enhancing cross-product realism.

6.7 Cross-Product Differentiation Validation

Product-level validation confirms that structurally distinct product designs produce **quantitatively and directionally consistent differences** in exposure, pricing, payment, affordability, risk, severity, and expected loss.

To preserve readability, product validation is shown in two layers: first, the **structure and affordability profile**, then the **risk and loss profile**.

6.7A Product Structure & Affordability Profile

Product	Applications	Avg Requested Amount	Avg APR	Avg Monthly Payment	Avg PTI
Mortgage	15,065	~$339K	8.75%	~$2,973	29.67%
Revolving Line	12,455	~$17K	24.68%	~$519	6.11%
Auto Loan	9,946	~$43K	11.18%	~$1,013	12.58%
Personal Loan	7,572	~$28K	15.70%	~$954	11.44%
HELOC	4,962	~$141K	11.86%	~$1,396	17.77%

The structure and affordability results confirm clear product separation. Mortgage generates the largest average exposure and highest average payment burden, while revolving lines carry the highest APR but much lower exposure and PTI. HELOCs provide a secured revolving contrast, with larger exposure than unsecured revolving lines but a lower immediate affordability burden than mortgage. Auto and personal loans occupy distinct installment profiles, with auto reflecting secured amortizing behavior and personal loans reflecting unsecured, higher-rate installment behavior. The revolving profile also reflects Workstream D: despite lower average exposure, payment behavior is now sensitive to APR movement, supporting the rate-to-affordability transmission path validated in Section 6.3.

6.7B Product Risk & Loss Profile

Product	Avg Estimated-PD proxy	LGD	Avg Expected Loss	Risk / Loss Interpretation
Mortgage	17.30%	20%	~$8,642	Low LGD but high EL due to large exposure
Revolving Line	15.33%	85%	~$2,143	Highest LGD; EL moderated by low exposure
Auto Loan	15.82%	45%	~$3,023	Moderate secured risk / severity profile
Personal Loan	16.19%	70%	~$3,048	Higher unsecured severity despite smaller balances
HELOC	17.13%	25%	~$5,911	Secured exposure with larger balances and lower severity

Risk and loss outputs vary appropriately by product. Although mortgage carries the lowest product-level LGD assumption, its larger exposure base produces the highest average expected loss, demonstrating that Expected Loss is driven by the interaction of exposure, estimated-PD proxy, and LGD rather than any single component alone. Revolving lines carry the highest severity assumption but lower average expected loss because exposure is much smaller. This confirms that the engine translates product structure into risk and loss outcomes in an economically coherent way.

6.8 Interpretation

Product-level validation confirms:

- Structural design differences translate into **meaningful behavioral variation**
- Affordability and risk respond to **product mechanics, not arbitrary assignment**
- The system supports **realistic portfolio composition and strategy testing**
- Risk gradients remain monotonic across score bands within each product, confirming that structural differentiation does not distort underlying credit-risk ordering.

6.9 Evidence Summary: Product Behavior Claims Supported

Claim	Evidence
Mortgage tail controlled	Max requested amount reduced ~$1.15M → ~$1.05M
Mortgage core preserved	Median mortgage requested amount remained ~$293K
Revolving APR sensitivity restored	Payment deltas positive across all revolving score bands
Scenario framework supports attribution	50,000 matched rows per scenario
Product differentiation retained	Distinct exposure, APR, payment, PTI patterns by product
PTI severe tail reduced	PTI > 50% declined from 1.77% to 1.20%
Estimated-PD proxy cap crowding reduced	Capped estimated-PD proxy observations declined from 17.72% to 12.86%

7. SCORE-BAND BEHAVIOR VALIDATION

7.1 Objective

Validate that credit score bands behave in a directionally realistic and analytically useful manner across borrower quality, affordability, behavioral risk, and modeled loss outcomes.

This section evaluates whether the engine produces a credible score-band gradient while preserving realistic overlap across the synthetic population. The objective is not to prove that the absolute estimated-PD values represent empirically calibrated default forecasts. Instead, the objective is to confirm that:

- Stronger score bands generally exhibit stronger borrower profiles.
- Weaker score bands generally exhibit higher behavioral and affordability risk.
- Estimated-PD proxy values increase directionally across score bands.
- Workstream B reduced mechanical compression at the upper estimated-PD cap.
- Expected Loss reflects the interaction of exposure, estimated-PD proxy, and LGD.

Important interpretation note:
The `estimated_pd` field in Module 1 is a **synthetic estimated-PD proxy**, not a production-calibrated default forecast. The 45% cap is a governed simulation ceiling designed to prevent runaway multiplier behavior, preserve a severe-risk region, and support scenario comparison. It should not be interpreted as an empirically observed or institution-calibrated default rate.

7.2A Observed Borrower & Affordability Score-Band Profile

Score Band	Applications	Avg Credit Score	Avg Income	Avg DTI	Avg Utilization	Avg PTI
Super Prime	8,984	805	~$169K	29.43%	20.38%	12.66%
Prime	13,987	730	~$143K	34.48%	29.94%	14.59%
Near Prime	12,073	670	~$118K	40.45%	42.37%	16.87%
Subprime	8,902	609	~$87K	46.57%	55.02%	18.84%
Deep Subprime	6,054	540	~$64K	52.98%	65.75%	22.09%

Interpretation

Borrower and affordability metrics move directionally as expected across score bands:

- Average credit score declines consistently from **805** in Super Prime to **540** in Deep Subprime.
- Average income declines from approximately **$169K** to **$64K**.
- Average DTI rises from approximately **29.43%** to **52.98%**.
- Average utilization rises from approximately **20.38%** to **65.75%**.
- Average PTI rises from approximately **12.66%** to **22.09%**.

This confirms that the engine produces an economically coherent borrower-quality gradient. Stronger score bands generally show stronger capacity and behavioral characteristics, while weaker score bands show higher debt burden, higher utilization, and higher payment stress.

The validation objective is not row-level determinism. Realistic portfolios contain overlap: some strong-credit borrowers are affordability-stretched, and some weaker-credit borrowers remain manageable on the requested credit. The aggregate score-band pattern confirms directional realism without artificially eliminating mixed-signal cases.

7.2B Observed Risk & Loss Score-Band Profile

Score Band	Avg Delinquencies	Major Derog Rate	Bankruptcy Rate	Avg Estimated-PD Proxy	Avg Expected Loss
Super Prime	0.07	0.57%	0.00%	1.36%	~$690
Prime	0.20	1.89%	0.13%	4.01%	~$1,807
Near Prime	0.66	4.51%	1.27%	13.09%	~$4,997
Subprime	1.58	10.59%	3.54%	35.93%	~$10,081
Deep Subprime	2.57	18.02%	5.63%	44.65%	~$9,549

Interpretation

Behavioral risk indicators worsen in the expected direction:

- Average delinquency count rises from **0.07** in Super Prime to **2.57** in Deep Subprime.
- Major derogatory rate rises from **0.57%** to **18.02%**.
- Bankruptcy rate rises from **0.00%** to **5.63%**.
- Average estimated-PD proxy rises from **1.36%** to **44.65%**.

This confirms that score band is functioning as the primary risk anchor, while behavioral and affordability variables provide additional differentiation.

The high estimated-PD proxy values in Subprime and Deep Subprime should be interpreted carefully. They represent the severe-risk end of a synthetic multiplier framework, bounded by a governed simulation cap. They are useful for validating ranking, segmentation, stress behavior, and expected-loss mechanics, but they are not calibrated production default forecasts.

One important nuance is that Subprime average expected loss is slightly higher than Deep Subprime average expected loss. This is not a defect. Expected Loss reflects the interaction of **exposure, estimated-PD proxy, and LGD**, not estimated-PD alone. Deep Subprime carries the highest modeled default-risk proxy, but lower average exposure can reduce average expected loss relative to Subprime. This is consistent with the engine's structure-to-risk design.

7.3 Estimated-PD Proxy Gradient Validation

The final V2.0 output confirms a clear and monotonic **estimated-PD proxy gradient** across score bands:

Score Band	Avg Estimated-PD Proxy
Super Prime	1.36%
Prime	4.01%
Near Prime	13.09%
Subprime	35.93%
Deep Subprime	44.65%

The progression is directionally correct and economically intuitive for a synthetic risk simulator: stronger score bands exhibit materially lower estimated risk, while weaker score bands exhibit progressively higher estimated risk.

However, the absolute values should not be interpreted as observed or institution-calibrated default rates. The Module 1 estimated-PD proxy framework is intentionally transparent and multiplier-based. It is designed to support:

- relative risk ordering,
- segmentation analysis,
- affordability stress testing,
- expected-loss comparison,
- and scenario sensitivity review.

It is not designed to replace empirical model development, calibration, back-testing, or production model validation.

Governance rationale for the 45% cap

The 45% cap functions as a **governed simulation ceiling**. Its purpose is to:

- Prevent runaway multiplier behavior in severe-risk combinations.
- Preserve a bounded severe-risk region without implying certainty of default or calibrated production default frequency.
- Maintain comparability across scenarios.
- Keep the model explainable and reviewable.
- Support stress-oriented portfolio analysis while avoiding mathematically extreme outputs.

The cap is therefore a modeling guardrail, not a forecast. The key V2.0 validation question is not whether 45% is an empirically perfect production threshold. The key question is whether the system avoids excessive mechanical compression at that threshold while preserving risk ordering.

Workstream B directly addressed that issue by reducing capped observations from **17.72%** to **12.86%**, improving high-risk differentiation while retaining the governance cap.

7.4 Affordability Alignment

Score-band behavior also supports the affordability design of the engine. Average PTI rises consistently as score quality weakens:

Score Band	Avg PTI
Super Prime	12.66%
Prime	14.59%
Near Prime	16.87%
Subprime	18.84%
Deep Subprime	22.09%

This validates that affordability pressure is directionally aligned with borrower risk profile while remaining structurally driven rather than mechanically score-driven.

PTI is influenced by product, amount, APR, term, payment logic, and income. Score band affects PTI indirectly through correlated borrower attributes and risk-sensitive pricing. That distinction is important: the engine should not assign affordability or risk solely from score. Instead, risk should emerge through the interaction of borrower quality, product structure, pricing, and affordability.

This result supports the central design chain:

Product Structure → APR → Payment → PTI → Estimated-PD Proxy → Expected Loss

7.5 Behavioral Risk Signal Validation

Behavioral variables also move in the expected direction:

Metric	Super Prime	Prime	Near Prime	Subprime	Deep Subprime	Validation Pattern
Avg Utilization	20.38%	29.94%	42.37%	55.02%	65.75%	Higher utilization in weaker bands
Avg Delinquencies	0.07	0.20	0.66	1.58	2.57	Delinquencies rise with risk
Major Derog Rate	0.57%	1.89%	4.51%	10.59%	18.02%	Severe derogatory signals concentrate in weaker bands
Bankruptcy Rate	0.00%	0.13%	1.27%	3.54%	5.63%	Bankruptcy remains rare but risk-concentrated

These patterns confirm that the engine does not rely on score alone. Score establishes the primary segmentation anchor, while utilization, delinquency, derogatory status, bankruptcy, DTI, file depth, and affordability variables reinforce risk directionality.

This is an important validation outcome because a credible synthetic credit engine must avoid both extremes:

- It should not generate variables independently with no coherent relationship.
- It should not force every variable to move mechanically in lockstep with score.

The observed results strike the intended balance: clear directional relationships with preserved variation and overlap.

7.6 Mixed-Signal Preservation

The score-band gradient is directionally strong, but the system does not force every borrower into a rigid profile. This is intentional.

The validation framework confirms that:

- Strong borrowers can still exhibit elevated PTI.
- Weaker borrowers can still show manageable PTI.
- Product structure can materially alter affordability even within the same score band.
- Expected Loss can vary by exposure and LGD even when estimated-PD proxy follows score-band ordering.
- High-risk observations remain bounded by governance controls rather than allowed to become unbounded.

This overlap is a sign of realism. Real credit portfolios contain exceptions, constrained but strong borrowers, and weaker borrowers with manageable new-payment burden. The engine preserves this behavioral complexity while maintaining clear average directional trends.

7.7 Conclusion

Score-band behavior in V2.0 is **directionally correct, economically coherent, and analytically useful**.

Validation confirms that:

- Borrower quality declines consistently from Super Prime to Deep Subprime.
- Affordability stress increases across weaker score bands.
- Behavioral risk signals worsen in the expected direction.
- The estimated-PD proxy increases monotonically across score bands.
- Workstream B reduced cap compression while preserving score-band risk ordering.
- Expected Loss reflects exposure, estimated-PD proxy, and LGD interaction rather than score alone.
- Realistic overlap remains present, avoiding artificial segmentation.

Overall, the score-band framework functions as intended: it anchors borrower risk while allowing product structure, affordability, behavioral variables, and loss severity assumptions to create realistic variation within and across segments.

The results support continued use of the estimated-PD proxy framework as a transparent synthetic risk proxy for strategy simulation, with the explicit boundary that it is not a calibrated production probability-of-default model.

8. AFFORDABILITY-TO-RISK RELATIONSHIP VALIDATION

8.1 Objective

Validate that affordability pressure, measured through Payment-to-Income ratio (PTI), translates into higher downstream estimated risk and Expected Loss in a directionally coherent and economically intuitive manner.

This section evaluates whether PTI functions as the intended bridge between loan structure and modeled risk outcomes. The objective is not to prove calibrated production default behavior, but to confirm that the synthetic engine preserves the intended relationship:

Payment Burden → PTI → estimated-PD proxy → Expected Loss

As established in the BRD, PTI is a central structural variable because it captures how requested credit burden interacts with borrower income. It is not assigned directly from credit score; rather, it emerges from product type, requested amount, APR, term, payment logic, and borrower capacity.

8.2 PTI Bucket Profile

The final V2.0 output confirms that higher PTI buckets are associated with higher payment burden, higher exposure, higher estimated-PD proxy values, and higher Expected Loss.

To preserve readability, the validation is shown in two layers:

1. **Structure and affordability profile**
2. **Estimated-risk and loss profile**

8.2A PTI Structure & Affordability Profile

PTI Bucket	Applications	% of Portfolio	Avg Monthly Payment	Avg Requested Amount	Avg APR
<5%	11,031	22.06%	~$334	~$16.5K	17.34%
5%–9%	11,846	23.69%	~$808	~$44.4K	16.78%
10%–14%	6,904	13.81%	~$1,206	~$79.4K	14.86%
15%–19%	4,076	8.15%	~$1,586	~$121.1K	13.77%
20%+	16,143	32.29%	~$2,938	~$304.5K	11.11%

Interpretation

The PTI structure profile confirms that affordability burden rises through economically intuitive channels:

- Average monthly payment increases from approximately **$334** in the lowest PTI bucket to approximately **$2,938** in the highest PTI bucket.
- Average requested amount increases from approximately **$16.5K** to approximately **$304.5K**.
- The highest PTI bucket is not driven by APR alone; in fact, average APR declines in higher PTI buckets because higher-burden observations are disproportionately influenced by larger secured balances and product structure.

This is an important validation point. The system is not mechanically assigning high risk to high APR accounts. Instead, affordability stress emerges from the interaction of **exposure, payment mechanics, product structure, and income capacity**.

8.2B PTI Estimated-Risk & Loss Profile

PTI Bucket	Avg Estimated-PD Proxy	Avg Expected Loss	Validation Interpretation
<5%	10.18%	~$648	Lowest affordability burden and lowest Expected Loss
5%–9%	12.09%	~$1,918	Manageable burden with modest risk increase
10%–14%	15.87%	~$3,365	Moderate burden with higher estimated risk
15%–19%	19.56%	~$5,137	Elevated burden with materially higher loss
20%+	23.02%	~$10,241	Highest affordability stress and highest Expected Loss

Interpretation

The estimated-risk and loss profile confirms that PTI functions as the intended affordability bridge:

- Average estimated-PD proxy increases from **10.18%** in the lowest PTI bucket to **23.02%** in the highest PTI bucket.
- Average Expected Loss increases from approximately **$648** to approximately **$10,241**.
- The estimated-PD proxy increases by approximately **2.3x** from the lowest to highest PTI bucket.
- Average Expected Loss increases by approximately **15.8x**, reflecting the combined effect of higher exposure, higher payment burden, higher estimated risk, and product-level LGD.

This confirms that Expected Loss is not driven by affordability alone. It reflects the modeled interaction of:

Exposure × estimated-PD proxy × LGD

The result is directionally coherent and consistent with the engine’s design objective: structural burden should flow into affordability stress, and affordability stress should contribute to downstream estimated risk and loss.

8.3 V2.0 Improvement Versus Prior Baseline

The PTI bucket relationship remained directionally intact after V2.0 enhancements. Importantly, V2.0 reduced high-burden severity without disrupting the core PTI-to-risk ordering.

High-PTI Bucket Metric	V1.0 Baseline	Final V2.0	Change	Interpretation
Avg monthly payment, PTI 20%+	~$2,979	~$2,938	↓ ~$41	High-burden payment pressure reduced
Avg requested amount, PTI 20%+	~$309.5K	~$304.5K	↓ ~$4.9K	High-burden exposure moderated
Avg estimated-PD proxy, PTI 20%+	23.64%	23.02%	↓ 0.62 pts	Estimated risk proxy modestly reduced
Avg Expected Loss, PTI 20%+	~$10,527	~$10,241	↓ ~$286	High-burden loss profile improved

Interpretation

The highest broad PTI bucket remained present, which is appropriate for realistic synthetic portfolio behavior. However, V2.0 reduced average payment burden, exposure, estimated-PD proxy, and Expected Loss within that high-burden bucket.

This is consistent with the Workstream A objective: improve affordability realism without eliminating valid high-burden cases. It also aligns with the tail-frequency evidence in Section 5.1, where the severe PTI tail declined from **1.77% to 1.20%** while the rare extreme tail remained preserved.

8.4 Scenario Sensitivity Confirmation

The affordability-to-risk relationship is also preserved under scenario variation.

In the final D1 200BP rate-up scenario:

Metric	Baseline	200BP Rate-Up Scenario	Change
Avg APR	14.56%	16.14%	+1.58 pts
Avg Monthly Payment	~$1,511	~$1,612	+$101
Avg PTI	16.50%	17.41%	+0.91 pts
Avg estimated-PD proxy	16.34%	16.39%	+0.05 pts
Avg Expected Loss	~$4,789	~$4,804	+$16

Additional row-level validation showed that **3,367 of 50,000 observations (~6.7%)** exhibited an increase in estimated-PD proxy under the rate-up scenario.

Interpretation

The scenario results confirm that affordability and estimated risk respond directionally to rate pressure:

APR ↑ → Payment ↑ → PTI ↑ → estimated-PD proxy ↑ → Expected Loss ↑

The effect is intentionally targeted rather than uniform. Most observations do not experience an estimated-PD proxy increase because the model requires the rate shock to flow through payment and PTI thresholds before affecting downstream estimated risk. This is consistent with the system's design: scenario changes should produce explainable, segment-sensitive effects rather than broad mechanical movement across all rows.

This also reinforces Workstream D. Revolving payment behavior now responds to APR movement, supporting the modeled transmission path while preserving the boundary that the revolving payment proxy is not a full billing-cycle model.

8.5 Validation Interpretation

Affordability-to-risk validation confirms four critical points:

1. **PTI is functioning as the intended bridge variable**
   Higher PTI buckets show higher average estimated-PD proxy values and higher Expected Loss.
2. **The relationship is economically coherent**
   Higher PTI reflects the interaction of payment burden, exposure, product structure, and borrower income.
3. **V2.0 improved the high-burden profile without suppressing realism**
   Severe PTI observations remain present, but high-burden payment, exposure, estimated risk, and loss levels were reduced.
4. **Scenario sensitivity is controlled and interpretable**
   Rate-up scenarios move APR, payment, PTI, estimated-PD proxy, and Expected Loss in the expected direction without creating population drift or uniform mechanical effects.

8.6 Conclusion

The affordability-to-risk relationship is validated.

Final V2.0 results confirm that:

- PTI increases with payment burden and exposure intensity.
- Higher PTI buckets show higher estimated-PD proxy values.
- Expected Loss rises materially across PTI buckets.
- V2.0 reduced high-burden severity while preserving realistic high-PTI observations.
- Scenario variation preserves the modeled transmission path from pricing into affordability and downstream estimated risk.

Overall, PTI is operating as the central affordability bridge intended by the system design. It links product structure and pricing to downstream estimated risk and Expected Loss in a transparent, directionally correct, and scenario-comparable manner.

9. WORKSTREAM VALIDATION NARRATIVE

9.1 Objective

Validate that each V2.0 workstream addressed a specific design or validation gap while preserving the stability, reproducibility, and interpretability of the overall simulation engine.

This section summarizes the iterative validation process used to evolve Module 1 from the validated V1.0 baseline into the final V2.0 architecture. Each workstream is evaluated using the following structure:

- Initial issue
- Why it mattered
- Change introduced
- Validation evidence
- Final outcome
- Residual interpretation

This workstream-based structure is important because V2.0 was not developed as a one-time model revision. It was refined through targeted validation cycles designed to improve realism, dispersion, comparability, and sensitivity without destabilizing the core synthetic population.

9.2 Workstream A — Mortgage Realism

Initial Issue

Initial behavior indicated that mortgage exposure and affordability tails required additional governance. Mortgage observations appropriately carried larger balances and higher payment burdens than other products, but validation showed that upper-tail exposure and payment behavior could still create elevated PTI outcomes.

This mattered because mortgage is the largest-balance product in the portfolio. If mortgage requested amount generation is not properly governed, it can distort:

- PTI distribution
- Expected Loss
- product-level affordability comparisons
- tail behavior interpretation
- scenario sensitivity

The goal was not to eliminate large mortgage balances or high-PTI observations. The goal was to ensure that mortgage tail behavior remained **plausible, bounded, and tied to borrower capacity**.

Iteration Summary

Iteration	Change Introduced	Validation Finding	Outcome
A1	Initial mortgage tail smoothing and diagnostic fields	Modest reduction in upper-tail exposure, but mortgage PTI remained elevated in certain high-balance cases	Improvement directionally correct, but insufficient
A2	Stronger amount-to-income tail smoothing	Further reduced maximum exposure and payment burden while preserving median mortgage behavior	Exposure tail improved, but PTI overlay still needed
A3 Final	PTI-aware affordability overlay added	Reduced upper-tail exposure and payment burden while preserving the core mortgage distribution	Accepted final mortgage realism design

Key Validation Evidence

Metric	V1.0 Baseline	A1	A2	Final A3	Interpretation
Portfolio max requested amount	~$1.15M	~$1.10M	~$1.05M	~$1.05M	Upper exposure tail controlled
Portfolio max monthly payment	~$10.3K	~$10.0K	~$9.45K	~$9.2K	Upper payment burden reduced
Mortgage avg requested amount	~$344K	~$344K	~$339K	~$339K	Tail reduced without broad compression
Mortgage median requested amount	~$293K	~$293K	~$293K	~$293K	Core mortgage distribution preserved
Mortgage avg monthly payment	~$3,017	~$3,012	~$2,976	~$2,973	Payment burden improved
Mortgage avg PTI	~29.82%	~29.96%	~29.69%	~29.67%	Central affordability stable

Final Outcome

Validation confirmed that Workstream A improved mortgage realism without overcorrecting the product.

The final A3 design:

- reduced upper-tail requested amount,
- reduced upper-tail monthly payment,
- preserved the median mortgage distribution,
- retained realistic high-balance mortgage observations,
- and introduced transparent diagnostic fields to support auditability.

The most important validation pattern is that **maximum values improved while medians remained stable**. This demonstrates that the change targeted the tail rather than compressing the entire mortgage population.

Residual Interpretation

High-PTI mortgage observations remain present, which is appropriate for a realistic synthetic portfolio. These cases are now interpreted as **bounded edge behavior**, not structural distortion.

This is consistent with the broader tail-frequency validation:

- PTI > 40% declined from **8.16% to 7.75%**
- PTI > 50% declined from **1.77% to 1.20%**
- PTI > 75% remained rare at **0.24%**

Workstream A therefore strengthened affordability realism while preserving valid edge cases for stress testing, manual-review realism, and scenario analysis.

9.3 Workstream B — Estimated-PD Proxy Dispersion

Initial Issue

Initial behavior indicated that high-risk observations were clustering too heavily near the estimated-PD proxy governance cap.

This mattered because excessive cap crowding reduces analytical usefulness. If too many high-risk observations collapse into the same capped value, the engine loses differentiation among weaker borrowers, severe-risk profiles, and high-affordability-stress cases.

The goal of Workstream B was not to lower risk artificially. The goal was to reduce mechanical compression while preserving:

- directional score-band ordering,
- severe-risk differentiation,
- bounded behavior,
- and the governance cap.

Iteration Summary

Iteration	Change Introduced	Validation Finding	Outcome
B1	Initial soft-saturation framework and diagnostic fields	Saturation zones worked, but **8,861 rows / 17.72%** still capped	Framework valid, but cap crowding remained high
B2	Tuned saturation behavior	Portfolio-level distribution improved modestly, but Deep Subprime remained close to cap	Further refinement required
B3 Final	Final soft-saturation calibration	Capped rows declined to **6,428 / 12.86%** while preserving score-band ordering	Accepted final estimated-PD proxy dispersion design

Key Validation Evidence

Metric	B1	B3 Final	Change	Interpretation
Total rows	50,000	50,000	No change	Population preserved
Capped estimated-PD proxy rows	8,861	6,428	-2,433 rows	Cap crowding reduced
% capped	17.72%	12.86%	-4.86 pts	Upper-bound compression materially reduced
Normal saturation zone	73.25%	73.25%	No change	Underlying risk-zone classification stable
High saturation zone	10.09%	10.09%	No change	Raw-risk segmentation preserved
Extreme saturation zone	16.66%	16.66%	No change	Severe-risk population preserved

Final Outcome

Validation confirmed that Workstream B improved estimated-PD proxy dispersion without weakening the risk framework.

The final B3 design:

- reduced mechanical compression near the governance cap,
- preserved the underlying saturation-zone structure,
- maintained score-band monotonicity,
- retained severe-risk behavior,
- and improved differentiation among weaker-credit observations.

The key validation insight is that the saturation-zone distribution did not change, but capped final observations declined materially. This means B3 did **not** redefine who was high risk. It improved how high-risk pressure translated into final estimated-PD proxy values.

Residual Interpretation

Some observations remain capped at the estimated-PD proxy governance maximum. This is intentional.

The cap is a **simulation guardrail**, not a production-calibrated default-rate ceiling. Remaining capped observations represent the severe-risk boundary of the synthetic framework. The V2.0 improvement is that fewer observations collapse mechanically at that boundary, improving differentiation while preserving bounded behavior.

9.4 Workstream C — Scenario Framework

Initial Issue

Initial versions of the engine produced deterministic synthetic populations, but did not yet include a formal scenario archive and matched comparison framework.

This mattered because a strategy simulator must support controlled comparison. Without formal scenario labels, archive persistence, and matched row-level comparison, users could inspect individual runs but could not reliably attribute differences across scenarios to specific parameter changes.

The goal of Workstream C was to transform Module 1 from a static synthetic population generator into a **scenario-capable comparison framework**.

Change Introduced

Workstream C introduced:

- `scenario_set_name`
- `scenario_name`
- archive persistence
- archive reset controls
- population anchoring through `population_id`
- portfolio-level scenario summaries
- product × score scenario summaries
- matched row-level delta outputs

Validation Evidence

Validation Check	Result	Interpretation
Baseline archived rows	50,000	Full population persisted
Rate-stress archived rows	50,000	Full comparison scenario persisted
Population ID	Same across scenarios	Borrower population held constant
Borrower-level attribute match	100% identical	No population drift
Matched row comparison	50,000 rows	Row-level attribution enabled
Scenario output deltas	Present and interpretable	Parameter changes produced measurable effects

Final Outcome

Validation confirmed that the scenario framework works as intended.

The framework enables:

- controlled baseline-versus-scenario comparison,
- direct row-level attribution,
- elimination of population noise,
- scenario archive auditability,
- and repeatable strategy testing.

This establishes Module 1 as a controlled experimentation framework rather than a one-run synthetic data generator.

Residual Interpretation

Scenario outputs may change in fields such as APR, payment, PTI, estimated-PD proxy, requested amount, and Expected Loss depending on scenario parameters and affordability logic. This is expected.

The validation requirement is not that all outputs remain identical across scenarios. The requirement is that **borrower-level attributes remain matched** and that outcome changes are attributable to controlled parameter differences. Workstream C met that requirement.

9.5 Workstream D — Revolving Sensitivity

Initial Issue

Initial scenario testing showed that revolving APR could change under rate-stress scenarios without producing corresponding movement in revolving payment burden.

This mattered because it broke an important transmission path:

APR → Payment → PTI → estimated-PD proxy → Expected Loss

If revolving payment remains fixed when APR changes, then rate scenarios understate affordability pressure for revolving products.

The goal of Workstream D was to restore directional APR sensitivity while preserving the modeling boundary that the revolving payment proxy is not a full billing-cycle or credit-card servicing engine.

Change Introduced

Workstream D introduced an APR-sensitive revolving payment proxy.

Instead of modeling revolving payment as a fixed minimum-payment percentage only, the final D1 design incorporates:

- a base minimum-payment component,
- an APR-sensitive interest-pressure component,
- bounded proxy behavior,
- and preservation of product-level simplicity.

This design restores rate sensitivity without attempting to model contractual card payment rules, statement balances, fees, grace periods, amortization schedules, or revolve behavior.

Before / After Evidence

In the prior scenario framework, revolving payment did not respond to APR movement.

Product	Score Band	Avg APR Delta	Avg Payment Delta
Revolving Line	Super Prime	+2.00%	$0.00
Revolving Line	Prime	+2.00%	$0.00
Revolving Line	Near Prime	+2.00%	$0.00
Revolving Line	Subprime	+2.00%	$0.00
Revolving Line	Deep Subprime	+1.58%	$0.00

After Workstream D, revolving payment deltas became positive across score bands.

Revolving Score Band	Avg APR Delta	Avg Payment Delta	Avg PTI Delta
Super Prime	+2.00%	+$15.84	+0.12 pts
Prime	+2.00%	+$15.00	+0.14 pts
Near Prime	+2.00%	+$14.25	+0.16 pts
Subprime	+2.00%	+$13.66	+0.21 pts
Deep Subprime	+1.58%	+$9.12	+0.20 pts

Portfolio-Level Scenario Evidence

Metric	Prior Scenario Framework	Final D1	Interpretation
Avg monthly payment delta	+$97.80	+$101.30	Stronger payment transmission
Avg PTI delta	+0.86 pts	+0.90 pts	Affordability impact restored
Avg estimated-PD proxy delta	+0.04 pts	+0.05 pts	Downstream estimated risk responds directionally
Avg Expected Loss delta	+$14.82	+$15.79	Loss impact slightly strengthened
Rows with estimated-PD proxy increase	3,036	3,367	More rows show downstream sensitivity

Final Outcome

Validation confirmed that Workstream D restored the missing revolving rate-transmission path.

The final D1 design demonstrates that:

- APR movement now affects revolving payment burden,
- payment burden flows into PTI,
- PTI contributes to downstream estimated risk,
- Expected Loss moves directionally,
- and the enhancement remains bounded as a simplified proxy.

This validates the intended chain:

APR ↑ → Payment ↑ → PTI ↑ → estimated-PD proxy ↑ → Expected Loss ↑

Residual Interpretation

The revolving enhancement is intentionally a proxy, not a full billing-cycle model.

It does not model:

- statement balances,
- minimum payment rule hierarchies,
- interest accrual timing,
- fees,
- transactor / revolver behavior,
- promotional APRs,
- or balance amortization.

This boundary is intentional and consistent with Module 1’s purpose as a strategy simulation engine rather than a product servicing model.

9.6 Integrated Workstream Conclusion

The V2.0 workstreams collectively improved the system without destabilizing the validated V1.0 foundation.

Workstream	Final Validation Outcome
A — Mortgage Realism	Upper-tail exposure and payment burden reduced while preserving core mortgage distribution
B — Estimated-PD Proxy Dispersion	Cap crowding reduced from **17.72% to 12.86%** while preserving score-band ordering
C — Scenario Framework	Matched row-level comparison validated across **50,000 rows per scenario**
D — Revolving Sensitivity	APR-sensitive payment response restored across all revolving score bands

Together, these enhancements confirm that V2.0 is:

- more realistic,
- more comparable,
- more diagnostically transparent,
- more sensitive to structural changes,
- and more useful for strategy testing.

Most importantly, the workstreams reinforce the system’s central design principle:

Risk outcomes are shaped by the interaction of borrower profile, product structure, pricing, affordability, and governed estimated-risk translation.

V2.0 therefore represents a controlled maturation of Module 1 from a validated synthetic population engine into a governed, scenario-capable decisioning simulation framework.

10. MIXED-SIGNAL REALISM VALIDATION

10.1 Objective

Validate that the synthetic portfolio preserves realistic mixed-signal borrower profiles rather than forcing applicants into rigid, deterministic risk segments.

The purpose of this section is to confirm that the engine produces credible exceptions, including:

- Strong-credit borrowers with elevated affordability burden
- Weaker-credit borrowers with manageable requested-payment burden
- Large-exposure borrowers with manageable affordability
- Product structures that materially alter affordability even within similar borrower profiles

This validation is important because real credit portfolios contain overlap. A realistic simulation engine should show clear average risk gradients while still preserving plausible edge cases and exceptions.

The objective is not to prove that every record is idealized. The objective is to confirm that exceptions are **present, explainable, bounded, and economically plausible**.

10.2 Validation Design

Mixed-signal realism was validated using targeted row-level diagnostic queries from the Section 9 QA framework.

These spot checks are designed to inspect whether realistic exceptions exist in the final population:

Diagnostic Test	Query Logic	Purpose
Strong credit but high PTI	Credit score ≥ 700 and PTI ≥ 18%	Confirms strong borrowers can still be affordability-stretched
Weaker credit but manageable PTI	Credit score < 640 and PTI < 10%	Confirms weaker borrowers are not uniformly overburdened
Large exposure but manageable affordability	Requested amount ≥ $250K and PTI < 15%	Confirms large exposure can be plausible when supported by income, term, and structure

These diagnostics are not intended to estimate portfolio prevalence. They are row-level plausibility checks. Portfolio-level tail frequency and distributional behavior are validated separately in Sections 5 and 8.

10.3 Mixed-Signal Evidence Summary

Mixed-Signal Test	Sample Evidence	Validation Interpretation
Strong credit but high PTI	Final B3 sample includes Prime mortgage borrowers with credit scores **719–759**, requested amounts approximately **$676K–$866K**, and PTI approximately **51.7%–55.1%**	Strong credit does not eliminate affordability stress when structure and payment burden are aggressive
Weaker credit but manageable PTI	Final B3 sample includes Subprime borrowers with credit score **639**, requested amounts approximately **$1K–$29K**, and PTI approximately **0.4%–5.9%**	Weak credit does not automatically imply high requested-payment burden
Large exposure but manageable affordability	Final B3 sample includes Super Prime mortgage borrowers with requested amounts approximately **$375K–$460K**, income approximately **$186K–$239K**, and PTI approximately **11.4%–14.9%**	Large exposure can remain manageable when supported by high income, longer term, and secured product structure

10.4 Strong Credit but High PTI

The strong-credit / high-PTI diagnostic confirms that stronger borrower profiles can still exhibit elevated affordability burden when product structure creates a large payment obligation.

In the final B3 sample:

Attribute	Observed Range / Profile
Sample size reviewed	25 records
Product type	Mortgage
Score band	Prime
Credit score range	719–759
Annual income range	~$164K–$202K
Requested amount range	~$676K–$866K
Monthly payment range	~$7.1K–$9.2K
PTI range	~51.7%–55.1%
Estimated-PD proxy range	~4.1%–9.8%

Interpretation

This diagnostic confirms that strong credit quality does not guarantee low affordability burden. The observed cases are driven primarily by mortgage structure: high requested amount, shorter term structure, high payment burden, and income interaction.

This is realistic. A Prime borrower may still request a large mortgage that creates elevated payment burden. The key validation question is whether those cases are explainable and bounded. The final B3 sample confirms that they are:

- The borrowers remain strong by score profile.
- The affordability burden is driven by large mortgage structure.
- The estimated-PD proxy remains materially lower than weak-credit severe-risk observations.
- The cases remain available for stress testing and manual-review realism.

This supports the broader conclusion that V2.0 did not eliminate high-PTI cases artificially. Instead, it governed the mortgage tail while preserving realistic affordability-stretched borrowers.

10.5 Weaker Credit but Manageable PTI

The weaker-credit / manageable-PTI diagnostic confirms that weaker borrowers are not automatically overburdened by the requested credit.

In the final B3 sample:

Attribute	Observed Range / Profile
Sample size reviewed	25 records
Score band	Subprime
Credit score	639
Product mix	Revolving Line, Unsecured Personal Loan, Auto Loan
Annual income range	~$49K–$139K
Requested amount range	~$1K–$29K
Monthly payment range	~$30–$581
PTI range	~0.4%–5.9%
Estimated-PD proxy range	~19.4%–45.0%

Interpretation

This diagnostic confirms that weak credit quality does not mechanically imply high requested-payment burden. In the sample, PTI remains manageable because requested amounts and payment obligations are relatively small.

At the same time, the estimated-PD proxy remains elevated because score band and behavioral risk signals continue to matter. This is exactly the intended behavior:

- Affordability can be manageable even when credit risk is elevated.
- Estimated risk should reflect more than PTI alone.
- Score, utilization, delinquency, file characteristics, and product structure all contribute to the estimated-PD proxy.

This confirms that the engine preserves separation between **credit risk** and **new-credit affordability**. That distinction is important for strategy testing because a borrower can be weak from a historical credit perspective while still carrying a manageable proposed payment.

10.6 Large Exposure but Manageable Affordability

The large-exposure / manageable-affordability diagnostic confirms that high requested amounts can remain plausible when supported by borrower capacity and product structure.

In the final B3 sample:

Attribute	Observed Range / Profile
Sample size reviewed	25 records
Product type	Mortgage
Score band	Super Prime
Credit score range	760–843
Annual income range	~$186K–$239K
Requested amount range	~$375K–$460K
Loan term	360 months
APR range	~5.65%–6.53%
PTI range	~11.4%–14.9%
Estimated-PD proxy range	~1.1%–1.7%
Expected Loss range	~$0.8K–$1.6K

Interpretation

This diagnostic confirms that large exposure does not automatically imply affordability stress. In this sample, higher income, Super Prime credit quality, 360-month term structure, and lower mortgage APR combine to produce manageable PTI despite large requested amounts.

This is an important realism check. A simulation engine that treated all large balances as high stress would distort mortgage behavior. Conversely, a simulation engine that allowed large balances without capacity support would create unrealistic exposure tails.

The final B3 output preserves the correct middle ground:

- Large balances remain present.
- They are tied to income capacity.
- Payment burden remains manageable.
- Estimated-PD proxy remains low.
- Expected Loss remains controlled despite meaningful exposure.

This supports Workstream A’s final outcome: mortgage tail behavior is governed, not eliminated.

10.7 Validation Interpretation

Mixed-signal validation confirms that the engine preserves realistic borrower overlap while maintaining coherent average gradients.

The final V2.0 output demonstrates that:

- Strong credit can coexist with high affordability burden.
- Weak credit can coexist with manageable new-payment burden.
- Large exposure can be affordable when supported by income, term, and pricing.
- Estimated-PD proxy behavior remains sensitive to borrower risk signals, not PTI alone.
- Expected Loss reflects the interaction of exposure, estimated-PD proxy, and LGD rather than any single factor.

This confirms that the engine avoids two common synthetic-data failure modes:

1. **Over-segmentation**
   Every strong borrower looks low-risk and every weak borrower looks high-burden.
2. **Random independence**
   Variables move without coherent relationship to borrower, product, affordability, or risk structure.

The final V2.0 behavior lands between those extremes: it preserves overlap while maintaining directional realism.

10.8 Conclusion

Mixed-signal realism is validated.

The final V2.0 output confirms that the engine produces credible exceptions and overlap across borrower quality, product structure, affordability, and estimated risk.

Validation confirms that:

- Mixed-signal cases exist in the final population.
- These cases are explainable through product structure, income, payment burden, and borrower characteristics.
- High affordability burden remains present but bounded.
- Weak credit does not automatically imply high PTI.
- Large exposure remains plausible when supported by income and term structure.
- The estimated-PD proxy remains directionally coherent and does not rely on PTI alone.

Overall, the engine successfully models realistic credit population complexity. It does not produce rigid stereotypes, and it does not generate disconnected random variables. Instead, it preserves the overlap, exceptions, and structural interactions necessary for credible pre-production strategy simulation.

11. EDGE CASE & BOUNDARY BEHAVIOR VALIDATION

11.1 Objective

Validate that extreme values, boundary behaviors, and residual edge cases in the final V2.0 output are **intentional, bounded, explainable, and non-destabilizing**.

This section evaluates whether the engine produces plausible edge behavior without introducing structural artifacts. The goal is not to eliminate all extreme observations. A realistic synthetic credit portfolio should contain tails, exceptions, and mixed-signal profiles. The validation objective is to confirm that those cases remain:

- logically explainable,
- limited in frequency,
- consistent with product and borrower structure,
- bounded by governance controls,
- and aligned with the intended use of the engine as a strategy simulation framework.

This section supports the BRD design principle that the system should model **realistic overlap and governed variation**, rather than producing rigid borrower stereotypes or idealized distributions.

11.2 Edge Case Validation Standard

Edge cases are considered acceptable when they meet the following criteria:

Validation Criterion	Interpretation
Explainable	The behavior can be traced to borrower attributes, product structure, affordability, pricing, or governed risk logic
Bounded	Values remain within defined model constraints, caps, or governed tail behavior
Non-dominant	Extreme cases do not materially dominate the portfolio or distort aggregate behavior
Economically coherent	The observation makes sense given income, exposure, payment burden, product type, and risk signals
Aligned to model boundaries	The behavior does not imply production underwriting, production pricing, calibrated default forecasting, or billing-engine functionality

Conversely, an edge case would be considered a validation concern if it showed:

- uncontrolled exposure scaling,
- broad concentration in extreme PTI levels,
- unexplained estimated-PD proxy cap crowding,
- impossible or structurally inconsistent product behavior,
- scenario movement caused by population drift,
- or outputs inconsistent with the system's intended simplified modeling boundaries.

11.3 Edge Case Evidence Summary

Edge Case / Boundary Area	Validation Evidence	Interpretation	Validation Outcome
High PTI frequency	PTI >40% declined from **8.16% to 7.75%**	High affordability stress remains present but limited	Acceptable / governed
Severe PTI frequency	PTI >50% declined from **1.77% to 1.20%**	Severe affordability stress materially reduced	Improved
Extreme PTI tail	PTI >75% remained rare at **0.24%**	Rare extreme tail preserved; not clipped away	Acceptable / realistic
Mortgage exposure tail	Max requested amount reduced **~$1.15M → ~$1.05M**	Upper exposure tail controlled	Improved
Payment burden tail	Max monthly payment reduced **~$10.3K → ~$9.2K**	Upper payment burden reduced	Improved
Estimated-PD proxy cap behavior	Capped observations reduced **17.72% → 12.86%**	Mechanical compression near governance cap reduced	Improved
Scenario sensitivity	**3,367 / 50,000** observations showed estimated-PD proxy increase under rate-up scenario	Scenario impact is targeted, not uniform	Acceptable / coherent
Mixed-signal profiles	Strong-credit/high-PTI, weak-credit/manageable-PTI, and large-exposure/manageable-affordability cases validated	Realistic overlap preserved	Acceptable / intentional
Revolving simplification boundary	APR-sensitive payment response restored, while full billing-cycle behavior remains out of scope	Proxy behavior works within intended boundary	Acceptable / bounded

11.4 Affordability and Exposure Tail Review

Affordability and exposure tails are the most important edge-case areas in Module 1 because they directly influence PTI, estimated-PD proxy behavior, and Expected Loss.

Validation confirms that V2.0 improved affordability tail behavior without eliminating realistic stress cases:

- PTI >40% declined from **8.16% to 7.75%**
- PTI >50% declined from **1.77% to 1.20%**
- PTI >75% remained rare and unchanged at **0.24%**

This pattern is important. V2.0 did not rely on arbitrary clipping or removal of edge cases. Instead, it reduced severe affordability stress while preserving rare extreme observations for realism, stress testing, and manual-review-style analysis.

Mortgage exposure behavior shows the same pattern:

- Maximum requested amount declined from approximately **$1.15M** to approximately **$1.05M**
- Maximum monthly payment declined from approximately **$10.3K** to approximately **$9.2K**
- Median mortgage requested amount remained stable at approximately **$293K**

This confirms that Workstream A controlled upper-tail behavior without collapsing the core mortgage distribution. The edge remained present, but the structural drivers of excessive tail inflation were reduced.

11.5 Estimated-PD Proxy Boundary Review

The estimated-PD proxy framework includes a governed upper cap. This cap is a **simulation guardrail**, not a calibrated production default-rate ceiling.

Validation confirms that Workstream B improved high-risk boundary behavior:

- Capped estimated-PD proxy observations declined from **8,861 rows / 17.72%** in B1
- to **6,428 rows / 12.86%** in final B3

This improvement reduced mechanical compression near the upper governance cap while preserving severe-risk observations.

The remaining capped observations are acceptable because:

- severe-risk cases should remain possible,
- the cap prevents runaway multiplier behavior,
- the cap preserves bounded scenario comparability,
- and final V2.0 reduced excessive clustering at the boundary.

The validation conclusion is not that the cap represents a calibrated default threshold. The conclusion is that the cap functions as a governed synthetic boundary, and that V2.0 improved differentiation below that boundary.

11.6 Scenario Boundary Review

Scenario validation confirms that edge behavior under scenario comparison is attributable to controlled parameter changes rather than population drift.

In the validated rate-up scenario:

- Baseline and rate-stress scenarios each retained **50,000 rows**
- Borrower-level attributes remained **100% identical**
- Scenario outputs changed only where parameter-driven logic propagated through pricing, payment, affordability, and estimated risk
- Approximately **6.7% of observations** showed an increase in estimated-PD proxy

This is an important boundary result. The rate scenario did not mechanically move every borrower. Instead, effects were concentrated in higher-risk and higher-PTI observations, consistent with model design.

This confirms that scenario edge behavior is:

- matched,
- attributable,
- directionally coherent,
- and not caused by stochastic population variation.

11.7 Product Boundary Review

The final V2.0 output also preserves product-specific modeling boundaries.

Most importantly, Workstream D restored APR-sensitive revolving payment behavior while preserving the intended simplification boundary. Revolving payment now responds directionally to APR movement, but the model does not attempt to replicate a full credit-card billing engine.

This distinction is important. The revolving payment proxy is designed to support strategy simulation by restoring the chain:

APR → Payment → PTI → estimated-PD proxy → Expected Loss

It is not intended to model:

- contractual minimum-payment hierarchies,
- statement balances,
- transactor versus revolver behavior,
- fees,
- grace periods,
- promotional rates,
- or full balance amortization.

Validation confirms that the simplified proxy is appropriate for Module 1's purpose: controlled pre-production strategy simulation, not servicing-system replication.

11.8 Residual Edge Behavior Interpretation

Residual edge behavior remains present in V2.0, but it is now properly interpreted.

The following residual behaviors are intentional:

- high-PTI observations remain present,
- rare extreme PTI cases remain preserved,
- some observations remain capped at the estimated-PD proxy governance maximum,
- large mortgage balances still exist,
- strong borrowers can still be affordability-stretched,
- weak borrowers can still have manageable requested-payment burden.

These behaviors are not validation failures. They are necessary to preserve realistic portfolio complexity.

The key validation finding is that these behaviors are now:

- explainable through the model design,
- bounded by governance controls,
- non-dominant in portfolio frequency,
- consistent with product and borrower structure,
- and appropriate for the system's strategy-simulation purpose.

11.9 Conclusion

Edge case and boundary behavior in V2.0 is validated.

Validation confirms that:

- affordability tails remain present but controlled,
- severe PTI frequency declined materially,
- mortgage exposure and payment tails were reduced,
- estimated-PD proxy cap crowding declined,
- scenario effects remain matched and attributable,
- product simplifications remain within documented boundaries,
- and mixed-signal borrower profiles remain realistic and explainable.

Overall, V2.0 does not achieve realism by removing edge cases. It achieves realism by governing them.

This supports the final validation conclusion that Module 1 operates as a **bounded, explainable, portfolio-realistic simulation engine** suitable for controlled strategy testing, sensitivity analysis, and decision framework design.

12. SYSTEM-LEVEL VALIDATION SYNTHESIS

12.1 Objective

Provide an integrated validation conclusion for Module 1 V2.0 based on the full validation evidence reviewed across portfolio structure, deterministic generation, scenario comparability, variable distributions, product behavior, score-band behavior, affordability-to-risk relationships, workstream outcomes, mixed-signal realism, and edge-case behavior.

This section evaluates whether the system, taken as a whole, satisfies its intended purpose:

To function as a **governed, explainable, scenario-capable, and portfolio-realistic synthetic credit simulation engine** suitable for pre-production strategy testing, sensitivity analysis, and decision framework design.

This conclusion is made within the documented system boundary: Module 1 is a strategy simulation and decision-support framework, not a production underwriting model, pricing engine, credit-card billing engine, or calibrated production probability-of-default model. The estimated-PD proxy is validated as a transparent synthetic risk proxy, not as an empirically calibrated default forecast.

12.2 Validation Evidence Summary

The final V2.0 validation evidence supports acceptance across the following dimensions:

Validation Dimension	Evidence Reviewed	Validation Conclusion
Portfolio structure	50,000 synthetic applications; product mix aligned within approximately **0.2 percentage points** across categories; score bands fully populated	Portfolio construction is stable and aligned to configured assumptions
Deterministic reproducibility	Repeated runs produced **100% identical outputs** under identical population and configuration controls	System is fully reproducible and audit-ready under identical inputs
Scenario comparability	Baseline and rate-stress scenarios each retained **50,000 rows**, same `population_id`, and **100% identical borrower-level attributes**	Matched row-level scenario comparison is valid
Variable distributions	Income, requested amount, PTI, and estimated-PD proxy distributions reviewed across central tendency and tail behavior	Distributions are realistic, bounded, and analytically usable
Tail governance	PTI >50% declined from **1.77% to 1.20%**; max requested amount declined **~$1.15M → ~$1.05M**; max payment declined **~$10.3K → ~$9.2K**	Severe affordability and exposure tails are improved and governed
Estimated-PD proxy dispersion	Capped estimated-PD proxy observations declined from **17.72% to 12.86%**	Upper-bound compression reduced while preserving score-band ordering
Product behavior	Product-level evidence confirmed distinct exposure, APR, payment, PTI, LGD, and Expected Loss profiles	Product structure materially influences affordability, estimated risk, and loss outcomes
Score-band behavior	Borrower quality, affordability stress, behavioral risk signals, and estimated-PD proxy values worsen directionally from Super Prime to Deep Subprime	Score-band behavior is directionally coherent and non-deterministic
Affordability-to-risk relationship	Higher PTI buckets showed higher average estimated-PD proxy values and materially higher Expected Loss	PTI functions as the intended affordability bridge into downstream estimated risk

Validation Dimension	Evidence Reviewed	Validation Conclusion
Workstream validation	Workstreams A–D were validated through targeted QA outputs, diagnostic evidence, and outcome analysis	V2.0 enhancements addressed known issues without destabilizing the core engine
Mixed-signal realism	Strong-credit/high-PTI, weaker-credit/manageable-PTI, and large-exposure/manageable-affordability cases were reviewed	Realistic overlap and exceptions are preserved
Edge-case behavior	Extreme values remain present but bounded, explainable, and non-dominant	Residual edge behavior is acceptable and consistent with portfolio realism

12.3 Design-to-Validation Traceability

The validation results directly support the design principles established for Module 1 V2.0.

Design Principle	Validation Evidence	Assessment
Deterministic Reproducibility	100% identical outputs under identical population and configuration controls	Satisfied
Full Auditability	Embedded Section 9 QA framework, scenario archive outputs, diagnostic fields, and matched row-level comparison	Satisfied
Parameter Governance	Product mix, score mix, rate environment, LGD assumptions, scenario configuration, archive controls, and Workstream D revolving payment assumptions are centrally parameterized and validated through the Section 3 fail-fast gatekeeper. The final V2.0 gatekeeper also enforces single-row parameter cardinality, ensuring that each run is governed by exactly one active configuration.	Satisfied
Separation of Concerns	Borrower attributes, product structure, affordability, estimated-risk proxy, LGD, and Expected Loss are generated in distinct stages	Satisfied
Scenario-Based Analysis	Scenario framework supports `scenario_set_name`, `scenario_name`, archive persistence, and matched row-level comparison	Satisfied
Behavioral Realism	Score-band gradients, mixed-signal cases, and edge behaviors remain realistic and non-deterministic	Satisfied

Design Principle	Validation Evidence	Assessment
Economic Coherence	APR, payment, PTI, estimated-PD proxy, and Expected Loss move directionally under scenario and product-structure changes	Satisfied

The validation evidence confirms that the system does not merely generate plausible-looking fields. It preserves the intended economic logic from product structure into affordability, affordability into estimated risk, and estimated risk into Expected Loss.

12.4 V2.0 Enhancement Acceptance Assessment

The final V2.0 workstreams are accepted for the intended simulation use case.

Workstream	Initial Validation Concern	Final V2.0 Outcome	Acceptance Assessment
A — Mortgage Realism	Mortgage upper-tail exposure and PTI behavior required stronger governance	Upper-tail exposure and payment burden reduced while median mortgage behavior remained stable	Accepted
B — Estimated-PD Proxy Dispersion	High-risk observations clustered excessively near the governance cap	Capped observations reduced from **17.72% to 12.86%**	Accepted
C — Scenario Framework	Scenario comparison required formal archive persistence and matched population controls	50,000 matched rows per scenario, same `population_id`, and row-level comparison validated	Accepted
D — Revolving Sensitivity	Revolving payment did not respond to APR movement under rate scenarios	APR-sensitive revolving payment response restored across all score bands	Accepted

These workstreams collectively improve realism, sensitivity, comparability, and auditability. Importantly, validation shows that the enhancements were **non-destructive** to the foundational V1.0 behaviors: product mix, score-band composition, deterministic population generation, and portfolio realism remained stable.

12.5 Residual Behavior Assessment

Residual edge behaviors remain present in V2.0. These include:

- high-PTI observations,
- rare extreme affordability cases,
- capped estimated-PD proxy observations,
- large mortgage exposures,
- strong-credit borrowers with elevated PTI,
- weaker-credit borrowers with manageable PTI,
- and product-specific structural variation.

These residual behaviors were reviewed and determined to be **intentional, bounded, and representative of real-world portfolio variability**.

They are not considered validation defects because:

- they are limited in frequency,
- they are explainable through product structure and borrower attributes,
- they remain within governed bounds,
- they do not destabilize aggregate portfolio behavior,
- and they support realistic strategy testing.

The system achieves realism by **governing edge behavior**, not by eliminating it.

12.6 Overall Validation Assessment

Based on the validation evidence reviewed, Module 1 V2.0 satisfies its stated validation objectives.

Validation confirms that the engine:

- produces stable and reproducible synthetic populations,
- preserves configured product and score-band structure,
- generates realistic borrower and product-level variation,
- maintains coherent affordability and estimated-risk relationships,
- supports matched scenario comparison,
- improves mortgage realism,
- reduces estimated-PD proxy cap compression,
- restores revolving APR sensitivity,
- preserves realistic mixed-signal cases,
- and maintains bounded, explainable edge behavior.

The system is therefore validated as a **portfolio-grade synthetic application and risk modeling engine** for its intended use: pre-production credit strategy testing, scenario analysis, sensitivity review, and decision framework design.

12.7 Acceptance Conclusion

Module 1 V2.0 is accepted as a **governed, explainable, scenario-capable, and portfolio-realistic simulation engine**.

The validation evidence supports continued use of the engine for:

- synthetic portfolio generation,
- product and score-band strategy analysis,
- affordability stress testing,
- estimated-risk proxy evaluation,
- Expected Loss comparison,
- matched scenario comparison,
- and pre-production decisioning framework design.

This acceptance is subject to the documented model boundaries. Module 1 V2.0 is not accepted or represented as:

- a production underwriting model,
- a production credit decision engine,
- a pricing optimization engine,
- a credit-card billing engine,
- a calibrated probability-of-default model,
- or a substitute for regulatory model validation.

Within its intended purpose and documented boundaries, Module 1 V2.0 has been successfully validated.

13. MODEL BOUNDARIES & INTENDED USE VALIDATION

13.1 Objective

Confirm that Module 1 V2.0 is validated for its intended purpose and that validation conclusions are interpreted within the correct system boundaries.

This section defines what the validation supports, what it does not support, and how results should be used by business, technical, and governance stakeholders.

Module 1 V2.0 is validated as a:

governed, deterministic, scenario-capable synthetic credit simulation engine

It is not validated as a production underwriting model, pricing model, billing engine, or empirically calibrated probability-of-default model.

This boundary is essential. The validation evidence confirms that the system is realistic, explainable, reproducible, and useful for strategy testing. It does not imply that the outputs should be used directly for real customer decisions or regulatory model production.

13.2 Validated Intended Uses

The final V2.0 validation supports use of the engine for the following purposes:

Intended Use	Validation Support	Interpretation
Synthetic portfolio generation	50,000-row deterministic population validated; product and score distributions aligned to configured assumptions	Suitable for controlled synthetic application-level portfolio creation
Pre-production credit strategy testing	Product, score-band, affordability, and estimated-risk relationships validated	Suitable for comparing strategy logic before implementation
Scenario analysis	Matched row-level scenario comparison validated using constant `population_id` and 50,000 rows per scenario	Suitable for controlled baseline-versus-scenario comparisons
Affordability sensitivity analysis	PTI bucket behavior and PTI-to-estimated-risk relationship validated	Suitable for evaluating how payment burden affects estimated risk and Expected Loss
Product mix / product structure analysis	Product-level exposure, APR, payment, PTI, LGD, and Expected Loss profiles validated	Suitable for analyzing structural differences across lending products
Estimated-risk proxy evaluation	Estimated-PD proxy ordering, dispersion, and cap behavior validated	Suitable for relative risk ranking and segmentation analysis
Expected Loss comparison	Expected Loss shown to respond to exposure, estimated-PD proxy, and LGD	Suitable for relative portfolio and scenario loss comparison
Audit and teaching artifact	Section 9 QA framework, deterministic design, diagnostic fields, and archive traceability validated	Suitable for explaining governed simulation design and validation logic

The system is therefore appropriate for **controlled analytical experimentation**, not operational decision execution.

13.3 Explicit Non-Uses

The validation does **not** support use of Module 1 V2.0 for the following purposes:

Non-Use	Reason
Production underwriting decisions	The system does not contain approved policy rules, real applicant data, or production decision governance
Customer-level credit approvals or declines	Synthetic outputs are not real customer records and should not drive actual credit decisions
Production probability-of-default modeling	`estimated_pd` is a synthetic estimated-PD proxy, not an empirically calibrated default forecast
Pricing optimization	APR is an affordability and scenario proxy, not an optimized offer price or profitability model
Credit-card billing or servicing simulation	Revolving payment is an APR-sensitive proxy, not a full billing-cycle or statement-balance model
Regulatory model validation	The framework is explainable and governed, but not a substitute for formal regulatory model validation
Fair lending or bias testing	The engine does not perform protected-class analysis, disparate-impact testing, or compliance adjudication
Macroeconomic forecasting	`rate_shift_bps` supports sensitivity testing, not macroeconomic forecast generation
Observed performance analysis	The system does not contain real booked-account performance histories, default outcomes, or time-series behavior

These exclusions are intentional and consistent with the system's design purpose.

13.4 Estimated-PD Proxy Boundary

The `estimated_pd` field is validated as a **synthetic estimated-PD proxy**, not as a production-calibrated probability-of-default model.

Validation confirms that the estimated-PD proxy:

- moves directionally with score-band risk,
- responds to behavioral risk signals,
- incorporates affordability stress through PTI,
- remains bounded by governed caps,
- shows improved high-risk dispersion after Workstream B,
- and supports Expected Loss comparison.

Validation does **not** confirm that the absolute estimated-PD proxy values represent observed default rates.

This distinction is critical. The estimated-PD proxy is designed for:

- relative risk ordering,
- segmentation,
- affordability sensitivity,
- scenario comparison,
- and Expected Loss simulation.

It is not designed for:

- empirical calibration,
- production model scoring,
- regulatory capital estimation,
- account-level default forecasting,
- or model deployment.

The 45% upper cap should therefore be interpreted as a **governed simulation ceiling**, not a calibrated production default-rate threshold. Workstream B validated that cap crowding was reduced from **17.72% to 12.86%**, improving differentiation near the upper boundary while preserving the severe-risk guardrail.

13.5 Scenario Framework Boundary

The scenario framework is validated for **matched, controlled comparison**, not for forecasting.

Validation confirms that:

- baseline and rate-stress scenarios each retained **50,000 rows**,
- scenarios used the same `population_id`,
- borrower-level attributes remained **100% identical** across compared scenarios,
- row-level deltas were attributable to parameter changes rather than population drift,
- and approximately **6.7% of observations** exhibited an increase in estimated-PD proxy under the rate-up scenario.

This supports controlled strategy testing.

It does not mean that the scenario output forecasts future market behavior. Scenario results should be interpreted as:

“What changes under this controlled assumption set?”

not:

“What will happen in the real world?”

The scenario framework is therefore valid for sensitivity analysis, not macroeconomic prediction.

13.6 Revolving Payment Boundary

Workstream D validated that revolving payment now responds directionally to APR movement. This restored the modeled transmission path:

APR → Payment → PTI → estimated-PD proxy → Expected Loss

However, the revolving payment logic remains a simplified proxy.

It does not model:

- statement balances,
- contractual minimum-payment hierarchies,
- transactor versus revolver behavior,
- fees,
- grace periods,
- promotional APRs,
- daily interest accrual,
- or full card amortization.

This simplification is intentional. Module 1 is a strategy simulation engine, not a credit-card servicing or billing engine.

Validation confirms directional APR-sensitive affordability behavior, not contractual billing accuracy.

13.7 Expected Loss Boundary

Expected Loss is validated as a **synthetic comparative loss metric**, calculated as:

Expected Loss = Exposure × estimated-PD proxy × LGD

Validation confirms that Expected Loss behaves coherently across:

- product types,
- score bands,
- PTI buckets,
- scenario deltas,
- and mixed-signal cases.

However, Expected Loss should not be interpreted as a production loss forecast. It is not calibrated to institutional loss history, macroeconomic loss curves, vintage performance, or observed recoveries.

Expected Loss is appropriate for:

- relative comparison,
- strategy tradeoff analysis,
- product-level contrast,
- scenario sensitivity,
- and portfolio simulation.

It is not appropriate for:

- financial reporting,
- regulatory capital modeling,
- reserve estimation,
- pricing optimization,
- or production profitability forecasting.

13.8 Validation Acceptance Boundary

The V2.0 validation supports acceptance of Module 1 for the following controlled use case:

Pre-production synthetic credit strategy simulation using governed, explainable, deterministic, scenario-comparable outputs.

The validation does not support unrestricted deployment.

The accepted interpretation is:

Valid Interpretation	Invalid Interpretation
“The engine produces realistic synthetic populations.”	“The engine produces real customer risk predictions.”
“Scenario deltas are attributable under controlled assumptions.”	“Scenario outputs forecast actual macro outcomes.”
“The estimated-PD proxy supports relative risk analysis.”	“The estimated-PD proxy is a calibrated production PD model.”
“Expected Loss supports comparative strategy testing.”	“Expected Loss is a production loss forecast.”
“Revolving APR sensitivity is directionally restored.”	“The system models credit-card billing behavior.”
“Tail behavior is bounded and realistic.”	“All edge cases should be eliminated.”

This boundary preserves the credibility of the validation conclusions.

13.9 Governance Requirements for Use

Any use of Module 1 V2.0 should follow these governance expectations:

- Parameter changes should be made through the centralized parameter layer.
- `population_id` should be held constant for matched scenario comparison.
- Scenario runs should be labeled using `scenario_set_name` and `scenario_name`.
- Section 9 QA outputs should be reviewed after material changes.
- Tail-frequency metrics should be monitored when changing amount, rate, or payment assumptions.
- Estimated-PD proxy cap behavior should be monitored when changing risk multipliers.
- Revolving assumptions should remain within documented proxy boundaries.
- Outputs should be interpreted as synthetic simulation results, not production forecasts.

These governance requirements ensure that the system remains reproducible, explainable, and audit-ready as additional scenarios are tested.

13.10 Conclusion

Module 1 V2.0 is validated for its intended purpose as a **governed synthetic strategy simulation engine**.

The validation supports use of the system for:

- controlled portfolio simulation,
- scenario comparison,
- strategy sensitivity testing,
- affordability analysis,
- estimated-risk proxy evaluation,
- and Expected Loss comparison.

The validation does not support use of the system as a production underwriting model, calibrated probability-of-default model, pricing optimization engine, credit-card billing engine, or regulatory model-validation substitute.

Within these boundaries, Module 1 V2.0 is fit for purpose. It provides a transparent, deterministic, auditable, and scenario-capable foundation for pre-production credit strategy evaluation.

14. FUTURE ENHANCEMENT ROADMAP

14.1 Objective

Define potential future enhancements that could extend Module 1 beyond its current validated V2.0 scope.

The roadmap items in this section are **not required for V2.0 acceptance**. Module 1 V2.0 has been validated for its intended purpose as a governed, deterministic, scenario-capable synthetic credit simulation engine.

Future enhancements are presented as optional extensions that could deepen analytical realism, expand downstream strategy testing, or support additional governance use cases in later modules or future versions.

This section distinguishes between:

- **validated V2.0 functionality**, which is accepted for current use,
- **intentional simplifications**, which are appropriate within Module 1's scope,
- and **future extensions**, which would require separate design, implementation, and validation.

14.2 Roadmap Framing

Module 1 V2.0 intentionally prioritizes:

- deterministic reproducibility,
- explainability,
- scenario comparability,
- product-aware structure,
- affordability-to-risk transmission,
- estimated-risk proxy behavior,
- and auditability.

The system does not attempt to solve every credit modeling problem in Module 1. Several areas remain intentionally simplified because they are outside the validated purpose of this module.

Future roadmap items should therefore be interpreted as **scope expansion opportunities**, not unresolved validation defects.

14.3 Roadmap Summary

Enhancement Area	Current V2.0 Boundary	Future Opportunity	Validation Implication
Decision strategy layer	Module 1 generates synthetic applications and risk outputs but does not execute production approval rules	Add policy, approval, decline, review, and line-assignment strategy logic in a downstream module	Validate approval rates, risk cuts, manual-review routing, and strategy tradeoffs
Empirical calibration / benchmarking	`estimated_pd` is a synthetic estimated-PD proxy, not a calibrated production default model	Compare proxy behavior to external benchmark ranges or institution-specific historical data where available	Validate calibration, discrimination, calibration drift, and absolute default-rate interpretation
Advanced scenario library	V2.0 supports baseline vs rate-stress scenario comparison	Add governed scenario library for product mix shifts, credit box changes, affordability thresholds, and macro sensitivity assumptions	Validate scenario attribution, reproducibility, and outcome explainability
Macroeconomic overlays	`rate_shift_bps` supports broad rate sensitivity only	Add unemployment, income stress, collateral value, inflation, or recession-style overlays	Validate macro-to-borrower transmission and avoid unsupported forecasting claims
LGD enhancement	LGD is product-level and intentionally simpler than the estimated-PD proxy framework	Introduce row-level LGD variation by collateral, exposure, LTV proxy, score band, or scenario	Validate severity differentiation and Expected Loss sensitivity

Enhancement Area	Current V2.0 Boundary	Future Opportunity	Validation Implication
Revolving behavior enhancement	Revolving payment is an APR-sensitive proxy, not a billing engine	Add utilization-sensitive balance behavior, statement balance proxies, or amortization-style behavior	Validate payment realism while maintaining simplified strategy-simulation boundaries
Lifecycle / performance simulation	Module 1 is application-level and does not model account performance time series	Add booked-account simulation, vintage behavior, delinquency transitions, or survival-style risk paths	Validate transition logic, timing, and portfolio performance realism
Fair lending / compliance analytics	Fair lending and protected-class analysis are out of scope	Add compliant testing framework if protected-class proxies or regulatory analysis are introduced	Requires separate compliance governance and fairness validation
Profitability / pricing layer	APR is an affordability proxy, not an optimized price	Add revenue, cost of funds, capital, operational cost, and risk-adjusted return logic	Validate profitability math and ensure pricing outputs are not misused
Governance reporting layer	QA outputs are available through SQL queries and validation documents	Add dashboards or automated validation reporting for scenario inventory and QA summaries	Validate report accuracy, lineage, and repeatability

14.4 Priority Roadmap Tiers

Future enhancements can be grouped into three practical priority tiers.

Tier 1 — Natural Next Extensions

These enhancements build directly on validated V2.0 capabilities and are the most logical next steps.

Priority Area	Rationale
Decision strategy layer	Module 1 already produces application-level borrower, product, affordability, estimated-risk, and Expected Loss outputs suitable for downstream strategy testing
Scenario library expansion	Workstream C validated matched scenario comparison, making the system ready for broader controlled scenario sets
Automated QA reporting	Section 9 already provides structured QA outputs; automation would improve repeatability and reviewer access
Parameter governance documentation	V2.0 is highly parameterized; a formal parameter dictionary would strengthen change control

Interpretation

Tier 1 enhancements are low-friction because they extend existing architecture rather than changing the core simulation engine.

Tier 2 — Analytical Deepening

These enhancements increase realism but require more careful validation.

Priority Area	Rationale
Row-level LGD refinement	Current LGD assumptions are intentionally product-level; row-level severity would improve Expected Loss nuance
Expanded macro sensitivity	Current macro sensitivity is rate-based; additional overlays could broaden strategy stress testing
Estimated-PD proxy calibration benchmarking	Proxy behavior is validated directionally; benchmarking could improve interpretability of absolute levels
Enhanced revolving behavior	Workstream D restored APR sensitivity; additional behavior could improve realism for revolving products

Interpretation

Tier 2 enhancements are valuable, but they introduce more modeling complexity. Each would require targeted validation to ensure the system remains explainable and governed.

Tier 3 — Advanced / Separate-Module Capabilities

These enhancements would materially expand the system beyond Module 1.

Priority Area	Rationale
Account performance simulation	Requires booked-account time-series logic and transition modeling
Survival / hazard modeling	Requires a fundamentally different modeling framework and validation approach
Fair lending analysis	Requires protected-class governance, compliance controls, and separate validation standards
Production underwriting deployment	Requires policy governance, model validation, monitoring, controls, and implementation review
Pricing optimization	Requires profitability, demand elasticity, compliance, and pricing governance

Interpretation

Tier 3 items should be treated as separate modules or separate project phases, not incremental edits to Module 1 V2.0.

14.5 Validation Requirements for Future Enhancements

Any future enhancement should follow the same validation discipline used for V2.0.

At minimum, future changes should include:

- clear statement of the initial issue or business objective,
- explicit modeling boundary,
- parameter documentation,
- deterministic reproducibility testing,
- regression-control validation against prior baselines,
- distributional validation,
- segment-level diagnostics,
- row-level spot checks,
- scenario comparison where applicable,
- edge-case review,
- and board-ready interpretation of results.

Future enhancements should also preserve the core V2.0 validation principles:

- **No uncontrolled population drift**
- **No unexplained tail expansion**
- **No unnecessary loss of mixed-signal realism**
- **No artificial over-compression**
- **No production-use claims without appropriate governance**
- **No interpretation of proxy outputs as calibrated forecasts unless separately validated**

14.6 Roadmap Governance Principles

Future work should follow these governance principles:

1. Preserve Module Boundaries

New functionality should not blur the distinction between:

- synthetic simulation,
- production underwriting,
- pricing optimization,
- billing logic,
- empirical default modeling,
- and regulatory validation.

If a future enhancement crosses one of those boundaries, it should be documented as a new module or separate validation scope.

2. Preserve Deterministic Comparability

Scenario testing should continue to preserve matched populations through `population_id`.

This enables:

- controlled comparisons,
- repeatable QA,
- row-level attribution,
- and audit traceability.

3. Preserve Explainability

Future enhancements should remain interpretable to business, risk, validation, and technical reviewers.

More complex logic should not be introduced unless it remains:

- documented,
- parameterized,
- testable,
- and explainable.

4. Preserve Validation Traceability

Every meaningful enhancement should include a validation narrative similar to the V2.0 workstream structure:

Initial issue → change introduced → validation evidence → final outcome → residual interpretation

This structure should remain the standard for future development.

14.7 Roadmap Interpretation

The roadmap confirms that Module 1 V2.0 is a strong foundation for future expansion.

The most natural next step is to build downstream decisioning modules that use the validated synthetic application, affordability, estimated-risk, and Expected Loss outputs generated by Module 1.

However, future expansion should remain disciplined. The system’s credibility comes from clearly separating what is validated from what is intentionally out of scope.

Module 1 V2.0 is validated for:

- synthetic application generation,
- product-aware structure,
- affordability analysis,
- estimated-risk proxy behavior,
- Expected Loss comparison,
- scenario testing,
- and governed pre-production strategy evaluation.

Future modules may extend that foundation, but should not retroactively reinterpret Module 1 as a production model.

14.8 Conclusion

The future enhancement roadmap identifies logical opportunities to extend Module 1 V2.0, but none are required for the current validation acceptance.

The validated V2.0 system is fit for its intended purpose. Future enhancements should be treated as additive extensions that require separate design, governance, and validation.

The recommended roadmap direction is:

1. build downstream decision strategy logic,
2. expand governed scenario libraries,
3. automate validation reporting,
4. enhance severity and macro sensitivity where appropriate,
5. and reserve production-grade modeling, compliance testing, pricing optimization, and account lifecycle simulation for separate modules or validation scopes.

This roadmap preserves the integrity of the current system while establishing a clear path for future enterprise-grade expansion.

15. FINAL VALIDATION STATEMENT

15.1 Final Determination

Module 1 — Synthetic Application & Risk Modeling Engine V2.0 has been successfully validated for its intended purpose.

The validation evidence confirms that the engine operates as a:

governed, deterministic, explainable, scenario-capable, and portfolio-realistic synthetic credit simulation engine

suitable for pre-production credit strategy testing, affordability analysis, product-structure evaluation, estimated-risk proxy analysis, Expected Loss comparison, and matched scenario review.

The system is accepted as fit for use within the documented Module 1 boundary.

15.2 Basis for Final Validation

The final validation conclusion is supported by evidence across all major validation dimensions:

Validation Area	Final Evidence
Portfolio construction	50,000 applications generated; product mix and score-band mix aligned to configured assumptions
Deterministic reproducibility	Repeated runs produced **100% identical outputs** under identical population and configuration controls
Scenario comparability	Baseline and rate-stress scenarios retained **50,000 matched rows per scenario** using the same `population_id`
Mortgage realism	Maximum requested amount reduced from approximately **$1.15M to $1.05M** while median mortgage requested amount remained approximately **$293K**
Affordability tail governance	PTI >50% declined from **1.77% to 1.20%**, while rare extreme PTI cases remained preserved
Estimated-PD proxy dispersion	Capped estimated-PD proxy observations declined from **17.72% to 12.86%**

Validation Area	Final Evidence
Revolving sensitivity	APR-sensitive revolving payment deltas became positive across score bands under rate-up conditions
Product differentiation	Products demonstrated distinct exposure, APR, payment, PTI, LGD, and Expected Loss profiles
Score-band behavior	Borrower quality, affordability stress, behavioral risk signals, and estimated-PD proxy values moved directionally across score bands
Mixed-signal realism	Strong-credit/high-PTI, weaker-credit/manageable-PTI, and large-exposure/manageable-affordability cases remained present and explainable
Edge-case behavior	Residual extremes were bounded, explainable, non-dominant, and consistent with realistic portfolio variability

Taken together, these results demonstrate that V2.0 preserves the validated V1.0 foundation while materially improving realism, sensitivity, comparability, and auditability.

15.3 Final Interpretation

The final V2.0 engine successfully demonstrates the core system thesis:

Credit risk in the simulation is shaped by the interaction of borrower profile, product structure, pricing, affordability, and governed estimated-risk translation.

The engine does not assign risk arbitrarily. It derives downstream estimated risk and Expected Loss through a transparent and explainable chain:

Rate Environment → APR → Payment → PTI → estimated-PD proxy → Expected Loss

Validation confirms that this chain behaves directionally as intended across product segments, score bands, PTI buckets, scenario runs, and edge-case profiles.

The V2.0 enhancements strengthen this chain by:

- governing mortgage exposure and affordability tails,
- reducing estimated-PD proxy cap compression,
- enabling matched row-level scenario comparison,
- restoring revolving APR-to-payment sensitivity,
- and adding diagnostic transparency for audit and review.

15.4 Acceptance Boundary

This validation supports use of Module 1 V2.0 as a synthetic strategy simulation and decision-support framework.

It does **not** support use of the engine as:

- a production underwriting model,
- a real customer credit decision engine,
- a calibrated probability-of-default model,
- a pricing optimization model,
- a credit-card billing or servicing engine,
- a macroeconomic forecasting model,
- a regulatory model-validation substitute,
- or a fair lending / compliance adjudication framework.

The `estimated_pd` field is validated as a **synthetic estimated-PD proxy**, not as a calibrated production default forecast. Expected Loss is validated as a comparative simulation metric, not as a production loss forecast.

These boundaries are intentional and consistent with the system's purpose.

15.5 Final Statement

Module 1 V2.0 has achieved its validation objective.

It produces a stable, reproducible, portfolio-realistic synthetic application population; preserves realistic borrower overlap and edge behavior; translates product structure into affordability; translates affordability into downstream estimated risk; supports Expected Loss comparison; and enables controlled, matched scenario analysis.

The final system is therefore validated as a:

portfolio-grade, governed, explainable, and scenario-capable synthetic credit decisioning simulation engine appropriate for pre-production strategy testing, sensitivity analysis, and decision framework design.

Within its documented boundaries, Module 1 V2.0 is accepted as complete and fit for purpose.

Final validation posture: Accepted for controlled synthetic strategy simulation within documented boundaries.